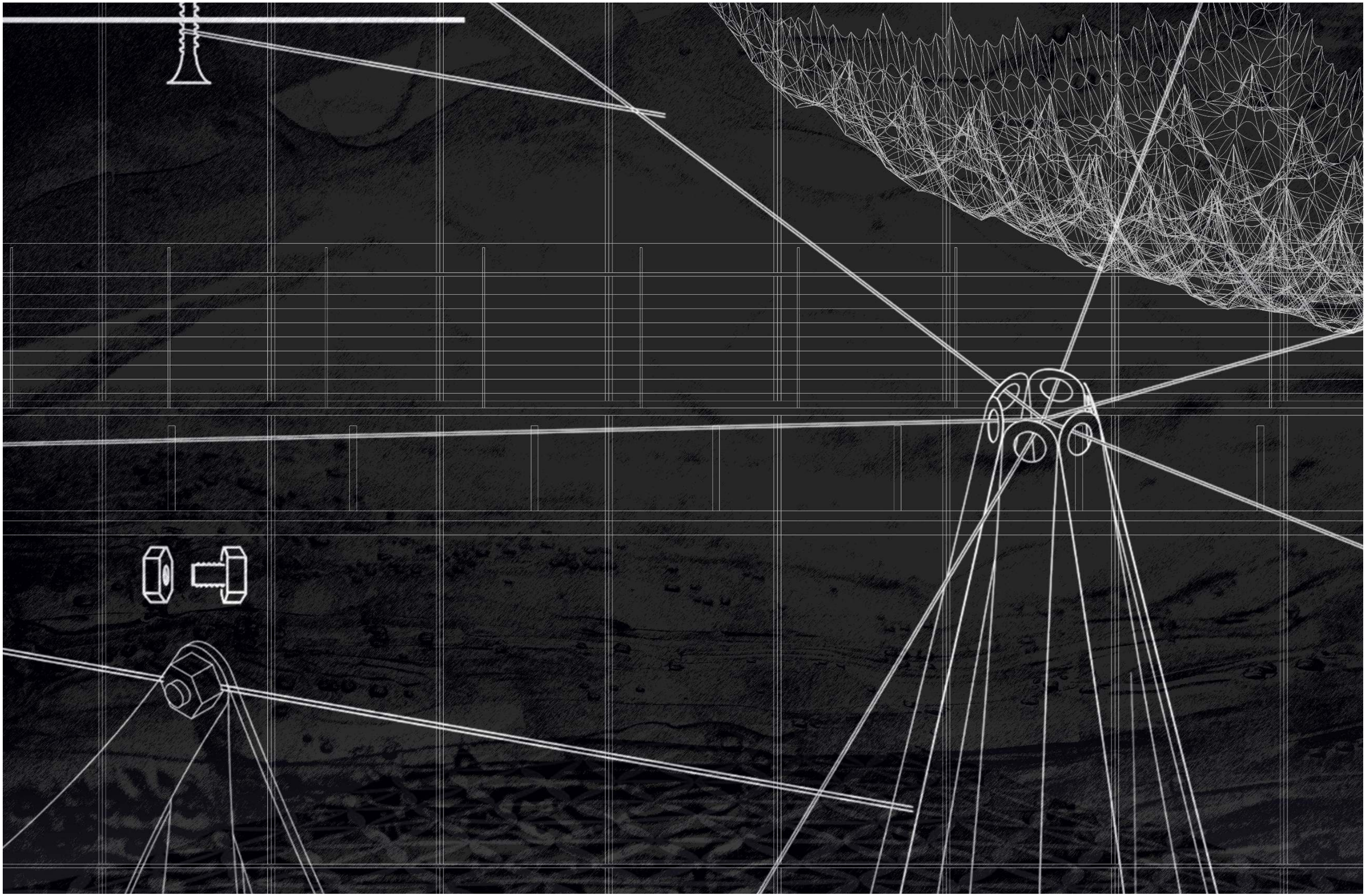
Daehong Kim

grizzlyraccoon@gmail.com
+82 010 2041 3863
Seoul, Republic of Korea



2013–2024

Portfolio.

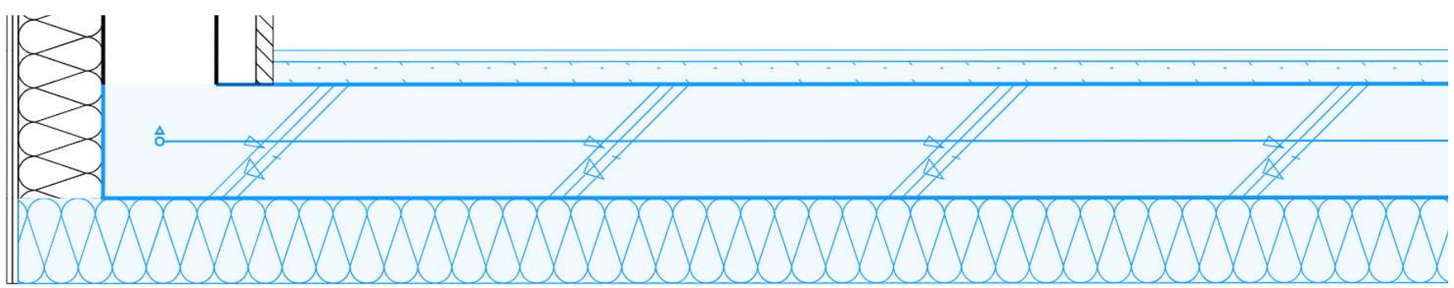
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Undergraduate Thesis Project

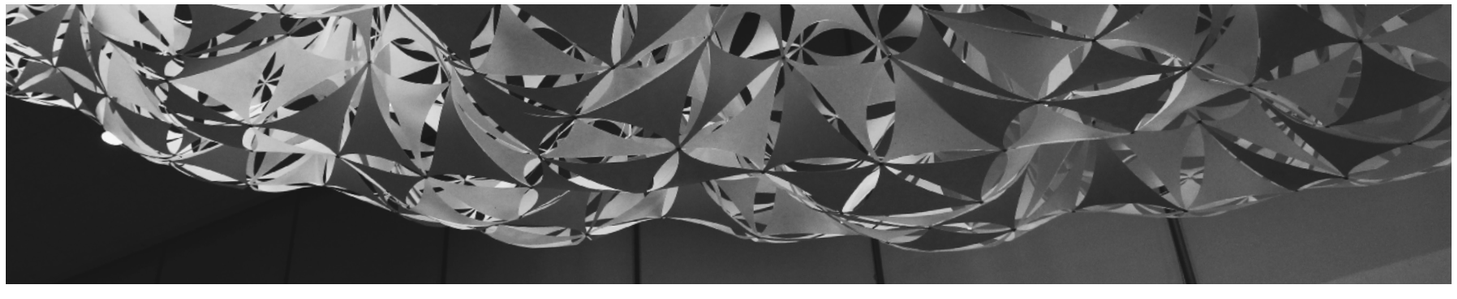
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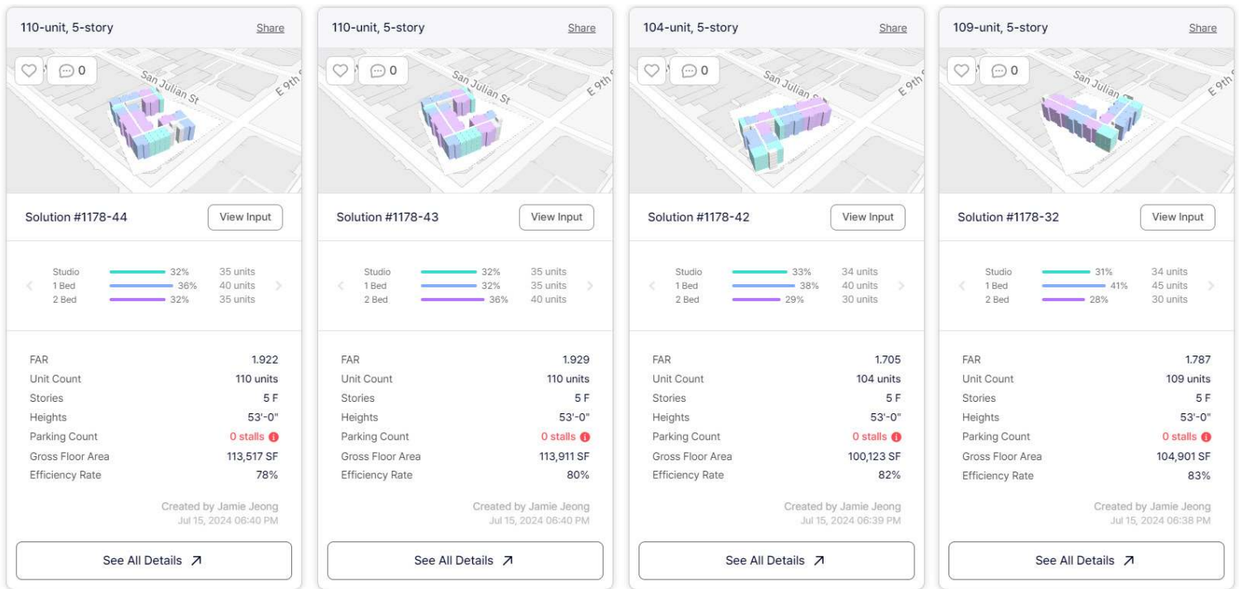
Zenerate App

Date	*2021 - 2024*
Type	*Professional Work*
Keywords	*Generative Plan Design, Optimization Algorithm, Zoning Constraints, Computational Geometry*
Office	*Zenerate*
Role	*Lead Computational Designer in a team of 4 to 5*
Tech Stack	*python, Node.js, mySQL, QGIS, git*

The Zenerate App is a cloud-based, automated feasibility study tool for multifamily modular housing development. To explore the potential of a site, the app generates multiple solutions to maximize constraints, including floor area ratio, density, and height regulations, while adhering to building codes and design preferences.

During the development of the Zenerate App, my role involved comprehensive development and management of the backend and core algorithms. Leading a team of four to five members responsible for floor plan generation algorithms, database design and management, code maintenance, and service business logic design, I took on hands-on development tasks while also overseeing and supervising the team's work.



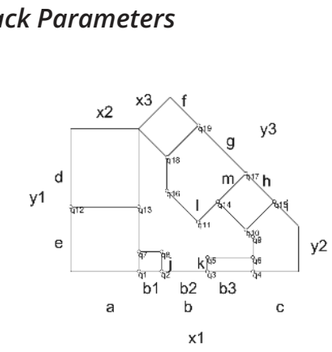


PlanMaker

This project originated in 2021 at Zenerate as a small-scale project focused on automating the creation of multifamily housing plans. The primary objective was to generate unit layouts that seamlessly adapted to a building's massing while maximizing leasable space. To achieve this, units were categorized according to several criteria, including dimensional ratios, outline shapes, corridor placement, number of bedrooms, and level of luxury. By systematically combining different variations from each category, the project produced hundreds of unique unit types without overcomplicating the underlying formation logic.

The foundation of PlanMaker's approach lay in a set of rule-based variables that filled each unit boundary automatically, ensuring the most efficient use of available space. Over time, these customizable parameters allowed for quick iterations and multiple design possibilities.

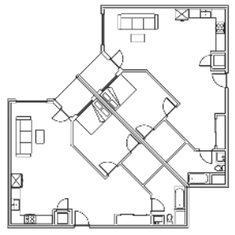
Pack Parameters

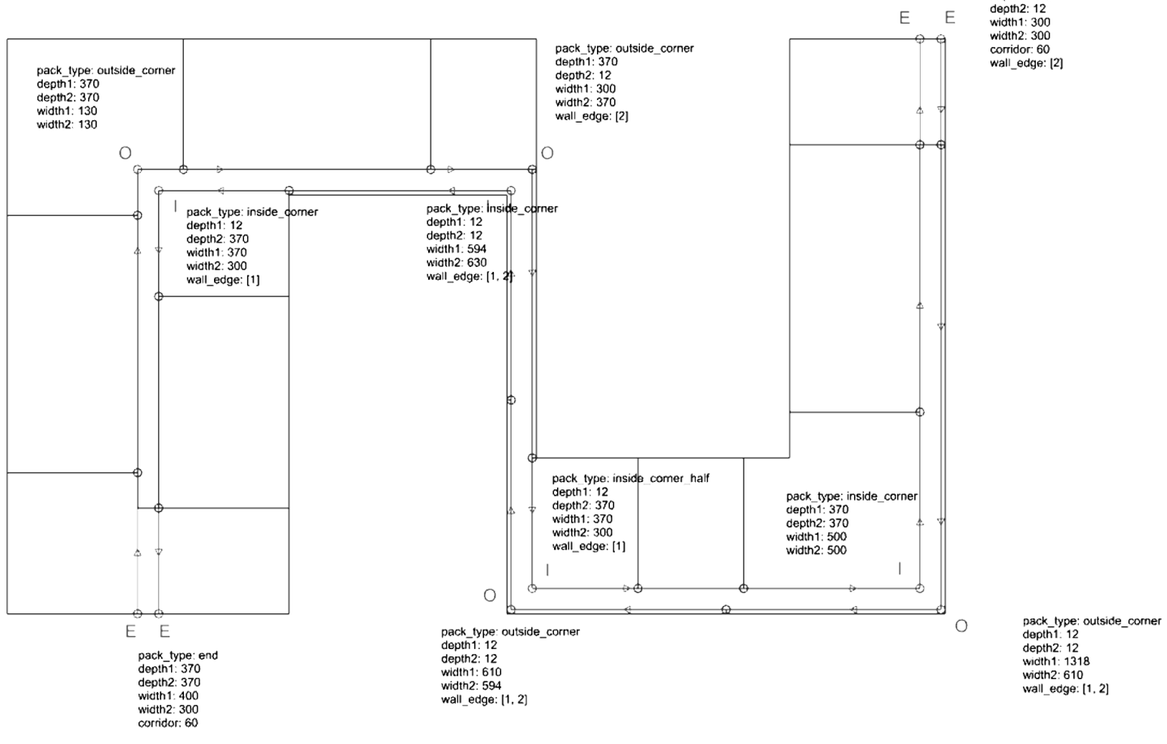


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pack_type: segment
wall_edge: None
layout: LA_onebed_r2
count: onebed: 1



pack_id: 11
pack_type: inside_corner
wall_edge: None
layout: LA_onebed_ic1, LA_onebed_ic1
count: onebed: 2





Zoning Code

Zenerate App's focus then turned to the city of Los Angeles with the aim of creating a comprehensive architectural plan for a multihousing building that strictly adheres to the local zoning code of a specific target site. The process began by gathering all relevant municipal zoning information and organizing it into a MySQL database. One of the key considerations identified by local developers was the Transit Oriented Communities (TOC) incentive, which grants benefits such as increased Floor Area Ratio (FAR) and reduced parking requirements for qualifying projects. However, the boundaries of TOC zones are determined by real-time transit schedules and can shift over time, making it necessary to establish a system that updates eligibility data as transit information changes.

To address this challenge, an automated workflow was set up to retrieve Los Angeles transit schedules through the city's public API. This data, which indicates where and when major transit stops operate, was then processed to determine which areas qualify for the various tiers of TOC incentives. The result was an evolving geospatial database that maintains an accurate map of TOC eligibility, reflecting any changes in local transit service. By frequently updating these data points and definitions, this database supports developers and stakeholders in navigating LA's complex zoning landscape and maximizing the potential of the TOC program in multihousing projects.



Transit Oriented Community Map of Los Angeles

Optimization

Zenerate App helps make architectural projects easier by gathering all important design and zoning details in one place. It collects information like property boundaries, setback rules, module sizes, and corridor widths. By keeping everything in a single system, Zenerate App ensures each design follows both the user's needs and the provided zoning regulations.

Once the inputs are set, Zenerate App creates a building layout that balances rules and design choices. Users can specify details such as how many floors they want, how many units they need, and whether to include extra modules or glazing. The app then combines these details to form a full layout, including the building's shape, floor plans, and circulation paths. It utilizes feedback loops to optimize key indices within a scheme.

The tool quickly generates multiple design options supporting many different building and massing types. It can also handle podium parking, ground parking, and even townhouse style projects with multiple buildings on the same site. This flexibility helps users find the best layout for each unique development scenario, while keeping to the regulations and user specific demands.



59 unit 5 story Podium Parking — Share

Solution #485-18 — View Input

Studio	32%	19 units
1 Bed	34%	20 units
2 Bed	34%	20 units

	Allowed or Required	Provided
FAR	Unlimited /	1.752
Unit Count	Unlimited /	59 units
Stories	Unlimited /	5 F
Heights	Unlimited /	53'0"
Parking Count	60 stalls /	57(+19) stalls
Gross Floor Area		94,253 SF
Efficiency Ratio		75%



59 unit 5 story Podium Parking — Share

Solution #485-17 — View Input

Studio	32%	19 units
1 Bed	34%	20 units
2 Bed	34%	20 units

	Allowed or Required	Provided
FAR	Unlimited /	1.742
Unit Count	Unlimited /	59 units
Stories	Unlimited /	5 F
Heights	Unlimited /	53'0"
Parking Count	60 stalls /	61(+11) stalls
Gross Floor Area		93,304 SF
Efficiency Ratio		76%

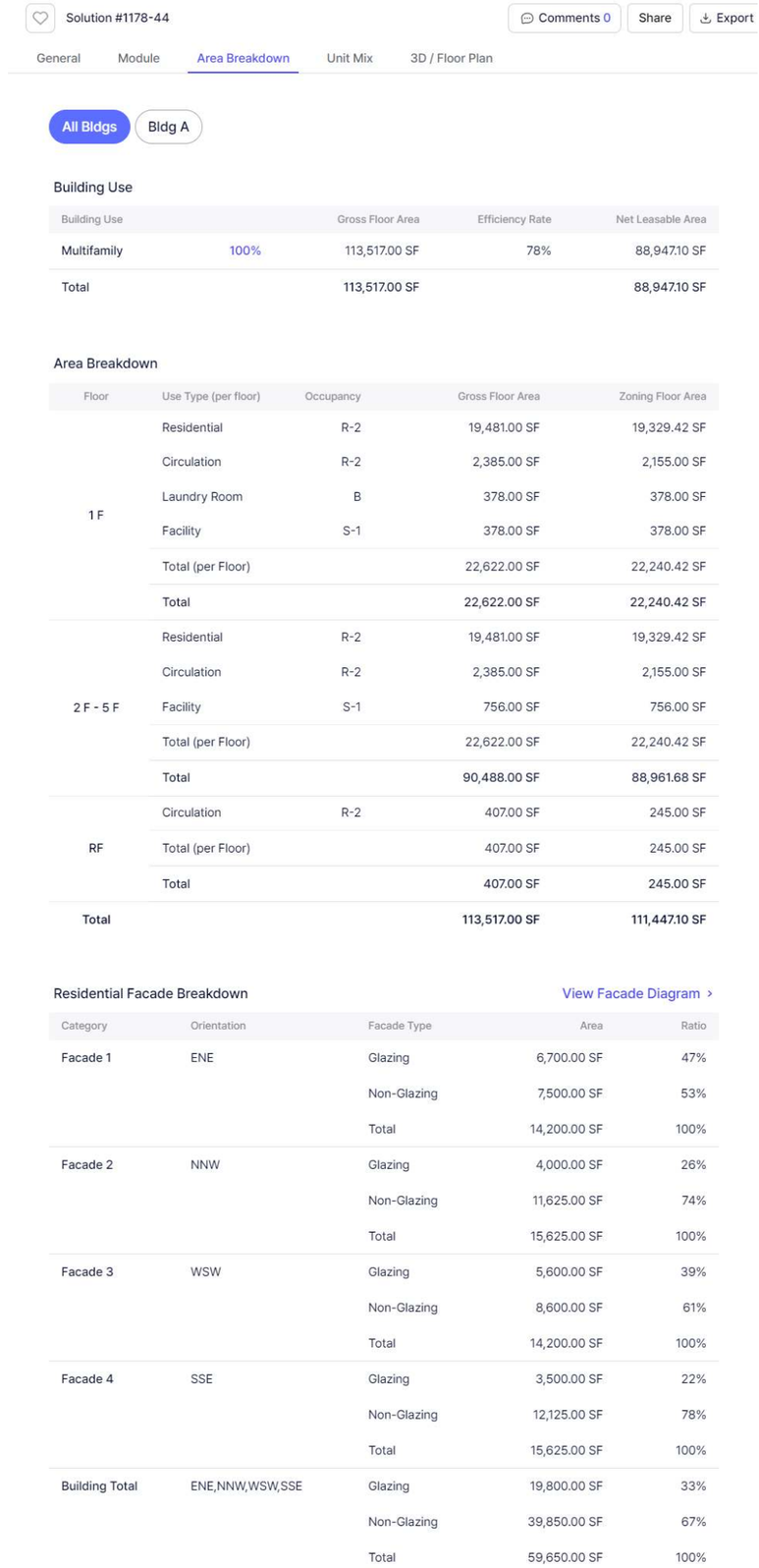


64 unit 5 story Podium Parking — Share

Solution #485-16 — View Input

Studio	33%	21 units
1 Bed	36%	23 units
2 Bed	31%	20 units

	Allowed or Required	Provided
FAR	Unlimited /	1.817
Unit Count	Unlimited /	64 units
Stories	Unlimited /	5 F
Heights	Unlimited /	53'0"
Parking Count	64 stalls /	63(+11) stalls
Gross Floor Area		97,223 SF
Efficiency Ratio		77%



Solution #1178-44 — Comments 0 — Share — Export

General | Module | Area Breakdown | Unit Mix | 3D / Floor Plan

All Bldgs | Bldg A

Building Use

Building Use		Gross Floor Area	Efficiency Rate	Net Leasable Area
Multifamily	100%	113,517.00 SF	78%	88,947.10 SF
Total		113,517.00 SF		88,947.10 SF

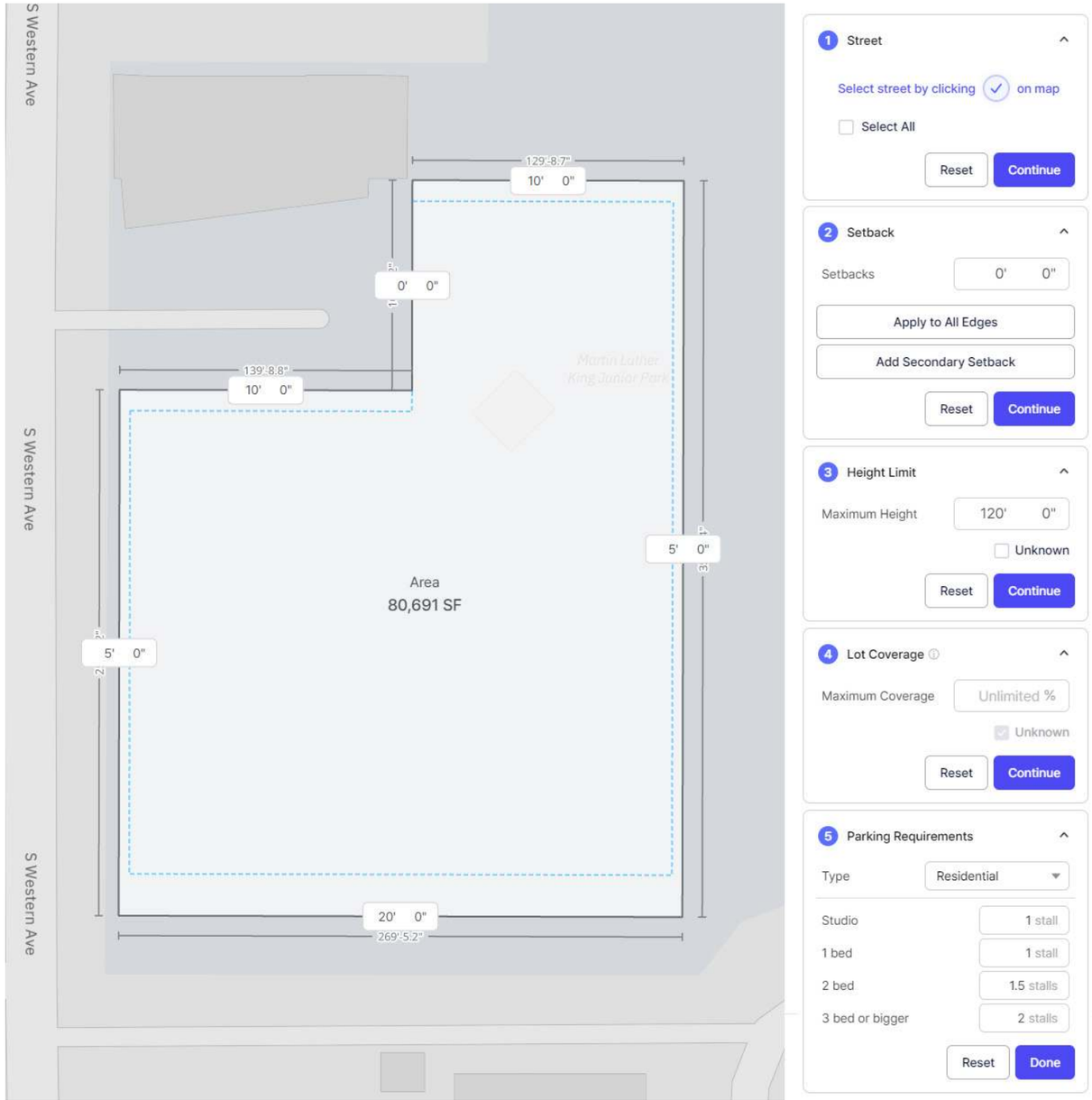
Area Breakdown

Floor	Use Type (per floor)	Occupancy	Gross Floor Area	Zoning Floor Area
1 F	Residential	R-2	19,481.00 SF	19,329.42 SF
	Circulation	R-2	2,385.00 SF	2,155.00 SF
	Laundry Room	B	378.00 SF	378.00 SF
	Facility	S-1	378.00 SF	378.00 SF
	Total (per Floor)		22,622.00 SF	22,240.42 SF
	Total		**22,622.00 SF**	**22,240.42 SF**
2 F - 5 F	Residential	R-2	19,481.00 SF	19,329.42 SF
	Circulation	R-2	2,385.00 SF	2,155.00 SF
	Facility	S-1	756.00 SF	756.00 SF
	Total (per Floor)		22,622.00 SF	22,240.42 SF
	Total		**90,488.00 SF**	**88,961.68 SF**
RF	Circulation	R-2	407.00 SF	245.00 SF
	Total (per Floor)		407.00 SF	245.00 SF
	Total		**407.00 SF**	**245.00 SF**
Total			**113,517.00 SF**	**111,447.10 SF**

Residential Facade Breakdown — View Facade Diagram ›

Category	Orientation	Facade Type	Area	Ratio
Facade 1	ENE	Glazing	6,700.00 SF	47%
		Non-Glazing	7,500.00 SF	53%
		Total	14,200.00 SF	100%
Facade 2	NNW	Glazing	4,000.00 SF	26%
		Non-Glazing	11,625.00 SF	74%
		Total	15,625.00 SF	100%
Facade 3	WSW	Glazing	5,600.00 SF	39%
		Non-Glazing	8,600.00 SF	61%
		Total	14,200.00 SF	100%
Facade 4	SSE	Glazing	3,500.00 SF	22%
		Non-Glazing	12,125.00 SF	78%
		Total	15,625.00 SF	100%
Building Total	ENE,NNW,WSW,SSE	Glazing	19,800.00 SF	33%
		Non-Glazing	39,850.00 SF	67%
		Total	59,650.00 SF	100%

Input Window



1 Street

Select street by clicking on map

Select All

Reset | Continue

2 Setback

Setbacks 0' 0"

Apply to All Edges

Add Secondary Setback

Reset | Continue

3 Height Limit

Maximum Height 120' 0"

Unknown

Reset | Continue

4 Lot Coverage

Maximum Coverage Unlimited %

Unknown

Reset | Continue

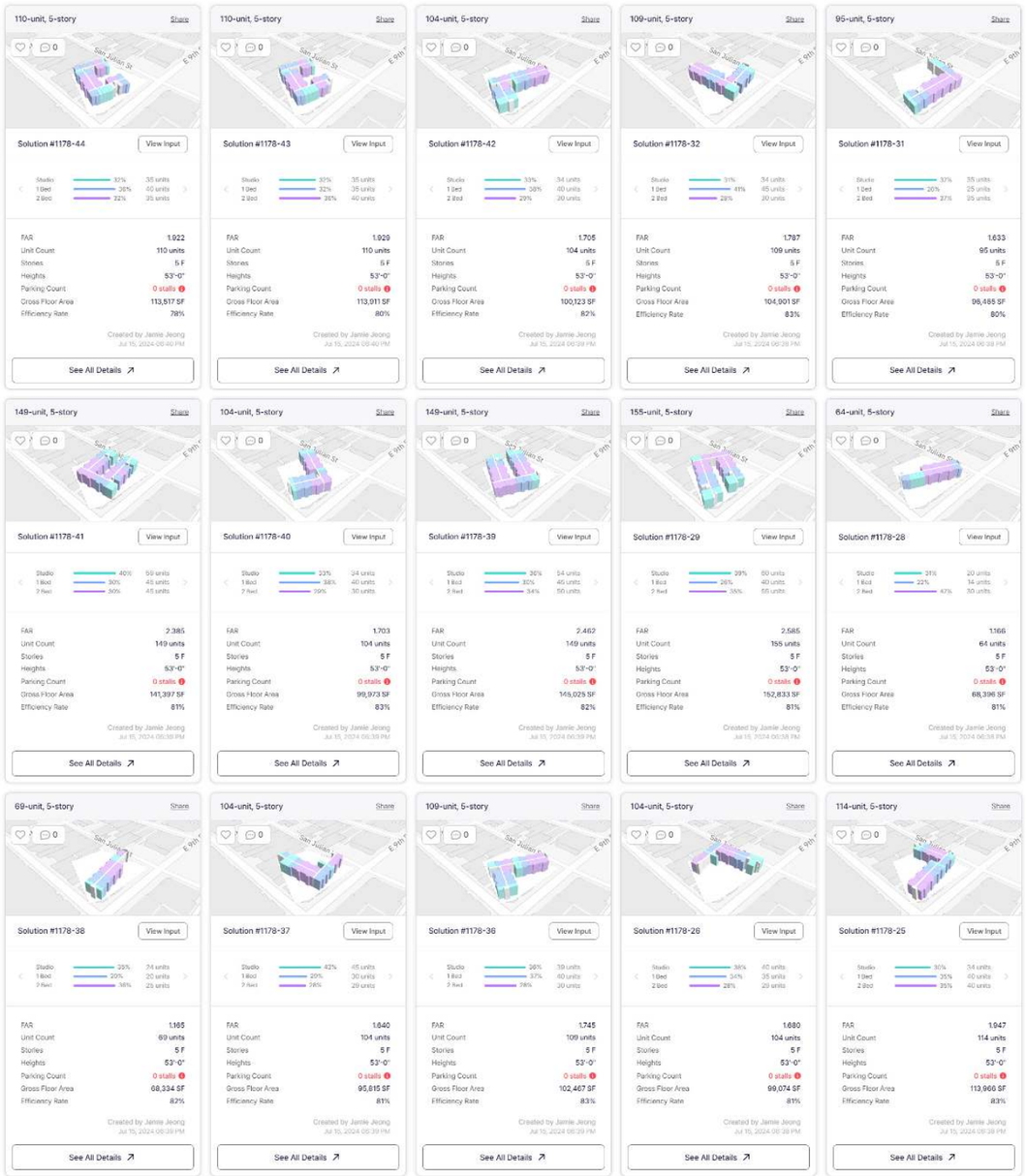
5 Parking Requirements

Type: Residential

Studio	1 stall
1 bed	1 stall
2 bed	1.5 stalls
3 bed or bigger	2 stalls

Reset | Done

Output Solutions



Modular Housing

To test the Zenerate App quickly and align it with market needs, a modular housing strategy was introduced. By applying modular logic, developers who design and sell modular units could use their own unit floor plans to generate multifamily apartment layouts directly within the platform. This holistic solution extended beyond residential units to encompass cores, amenities, and other non-residential elements, guaranteeing that every aspect of the building could be produced in a modular format. As a result, users gained the ability to visualize exactly how their various modules interconnect through a detailed modular diagram, providing valuable insights into design feasibility and efficiency.

In collaboration with SoLa Impact, a leading developer of affordable housing in California, I led the development and communication efforts to customize the Zenerate App for their in-house designed modular unit library. This partnership enabled SoLa Impact to integrate their proprietary modular designs into the platform, streamlining the planning and configuration of multifamily apartment projects. By tailoring the software to accommodate SoLa Impact's specific module types, the Zenerate App facilitated the efficient generation of scalable, cost-effective housing solutions. This customization not only supported SoLa Impact's mission to provide high-quality, sustainable affordable housing in Los Angeles but also demonstrated the versatility and adaptability of the Zenerate App in meeting diverse real estate development needs.

Sample Modules



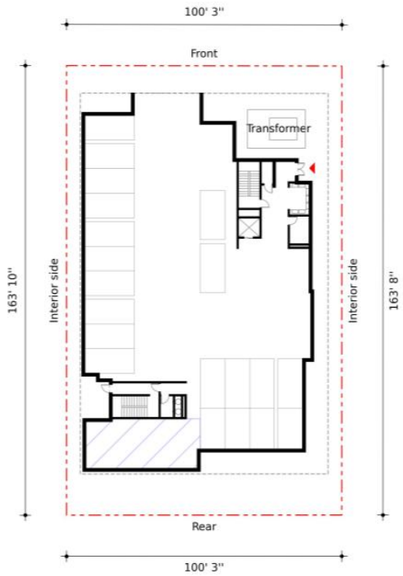
SoLa Impact Modular Output

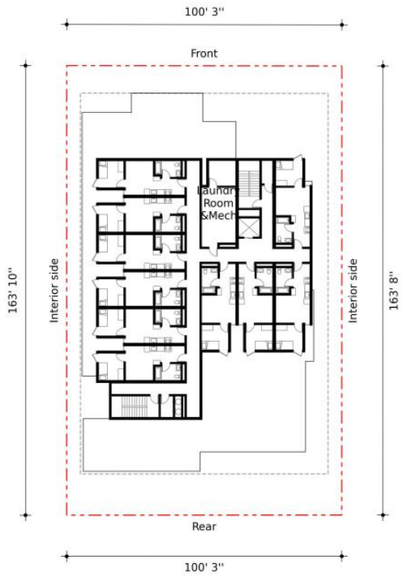
Provided by Zoning

	Allowed	Provided
Density	21 units	21 units
ZFA	37,551 SF	11,271 SF
Story	6 F	3 F
Height	45 ft	31.5 ft

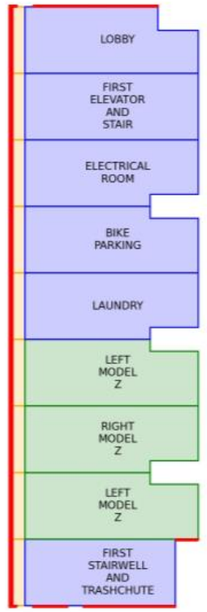
Project Information

Building Efficiency	64.23%
Residential Module Count	21
Non-res Module Count	5





Modular Diagram



CORRIDOR: 9
LEFT_MODEL_Z: 2
FIRST_ELEVATOR_AND_STAIR: 1
LOBBY: 1
FIRST_STAIRWELL_AND_TRASHCHUTE: 1
ELECTRICAL_ROOM: 1
RIGHT_MODEL_Z: 1
BIKE_PARKING: 1
LAUNDRY: 1

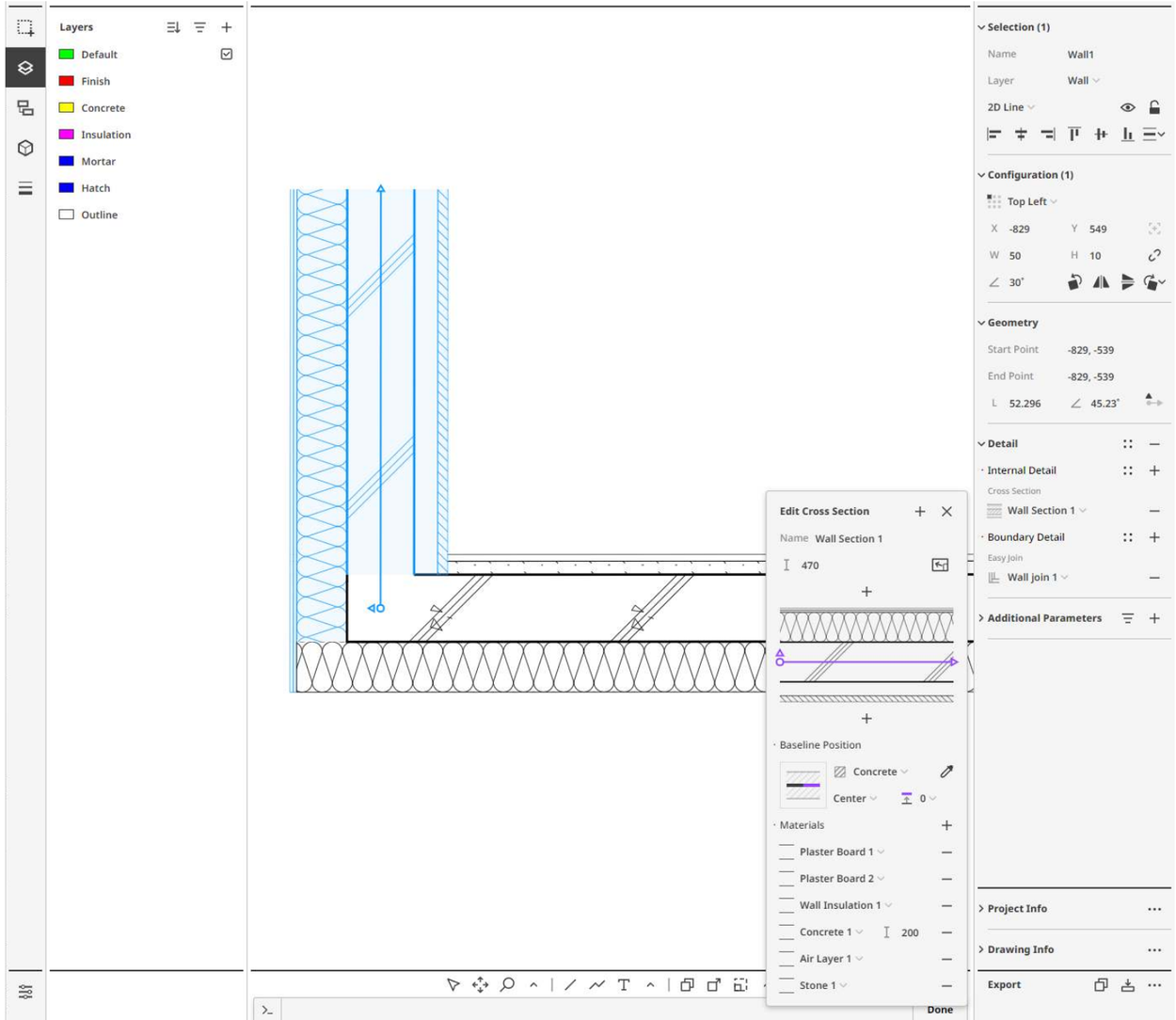
SOLA IMPACT

zenerate

Joint

Date	*October, 2024 - present*
Type	*Personal*
Keywords	*Modelling Software* *Web Development* *BIM*
Collaborators	*Changyong Kim*
Role	*All stages of development shared between team of two.*
Tech Stack	*Node.js, React.js, Three.js, MongoDB, git*

This experimental web-based service aims to streamline architectural detailing by automating and customizing section details through a simple skeleton-line approach. With each new line drawn, joint details are created instantly, and a suite of intuitive editing tools allows users to set priorities, reorder materials, and fine-tune baselines. Although still in progress, the project's future milestones include building a comprehensive database of section details for users to select and customize, as well as integrating an AI model to suggest details based on verbal prompts.

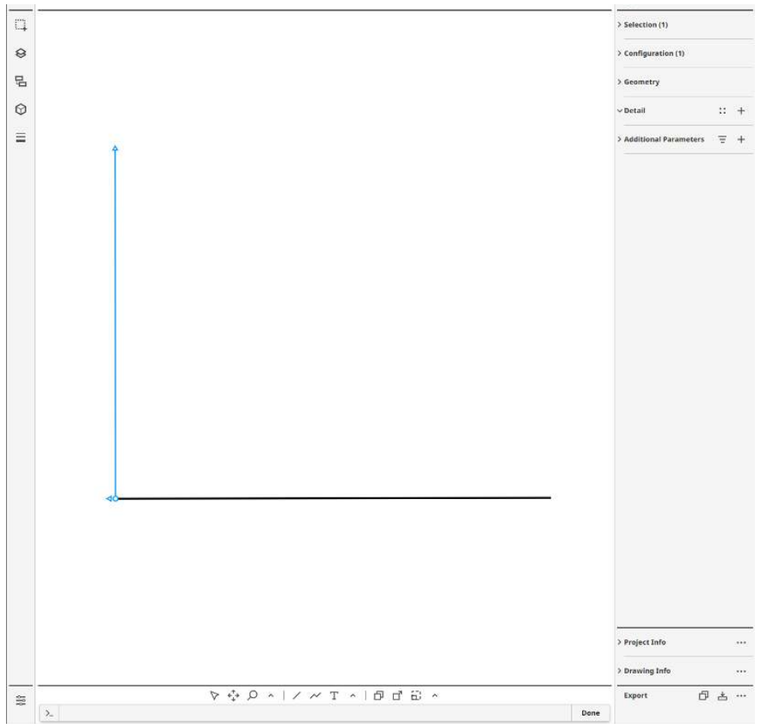


Automated Detail Generation

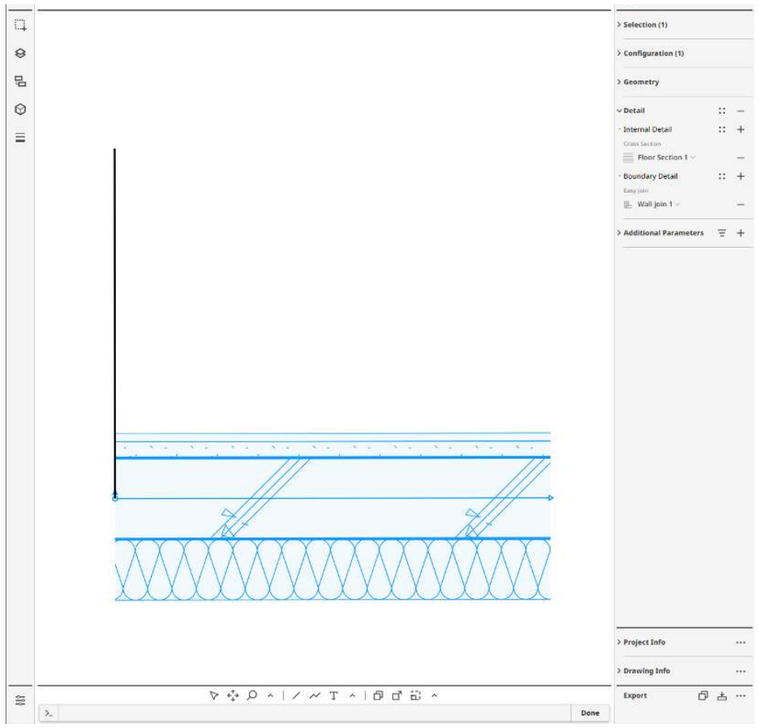
Central to this tool is the ability to translate a single skeleton line into detailed section connections. Whenever a user draws a new line intersecting an existing one, the system automatically generates a joint detail based on the designated materials. This eliminates much of the repetitive drafting typically associated with construction documentation, enabling architects and designers to focus on conceptual and design-oriented tasks.

Material Editing & Configuration

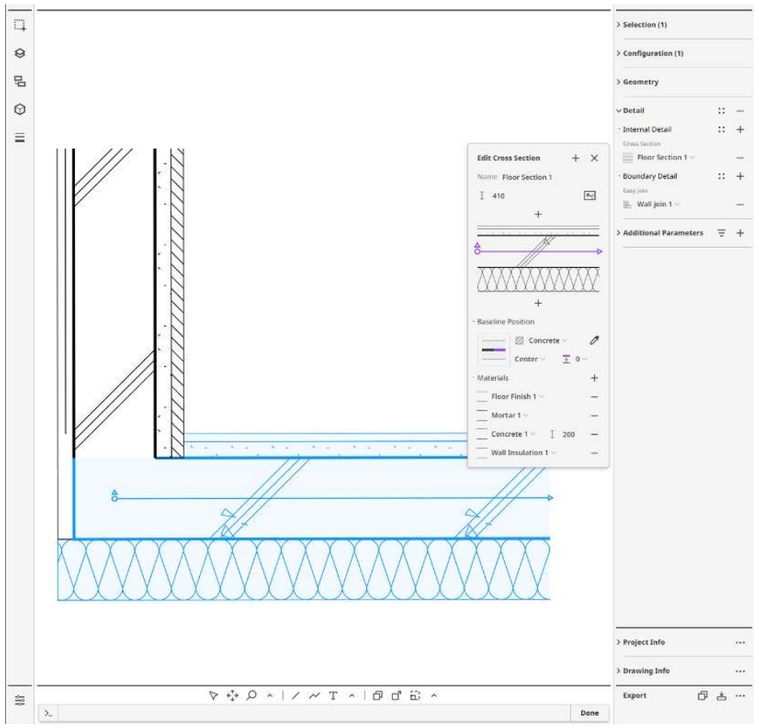
A key feature of the platform lies in its customizable material management. Users can alter the priority of materials when they overlap, ensuring the correct visual or functional hierarchy for each junction. Reordering materials is as simple as dragging them in a list, immediately updating their positions in the detail. For more nuanced edits, any material layer can be changed or replaced, giving full control over the final composition of each section detail.



Line Drawing



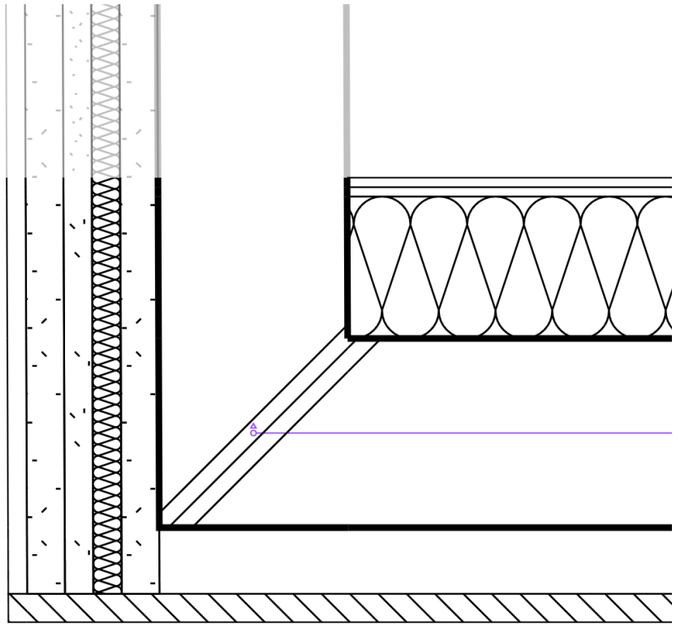
Section Generation



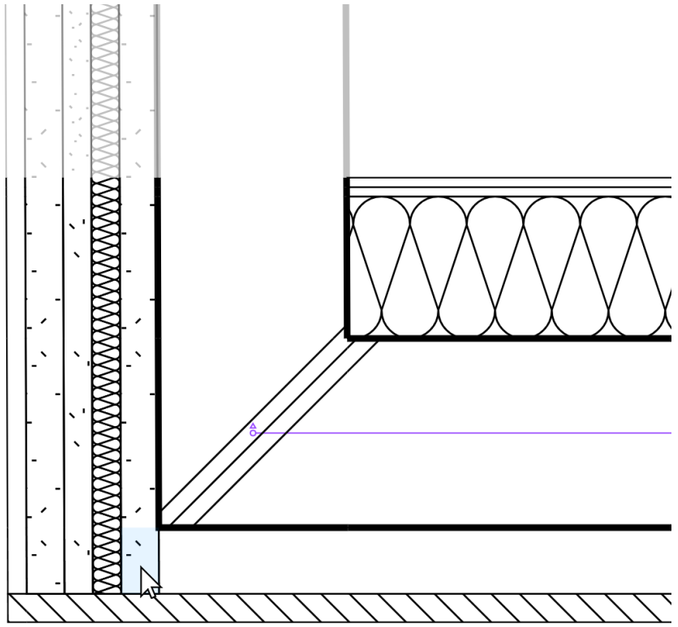
Section & Joint Editing

Joint Detail Editing

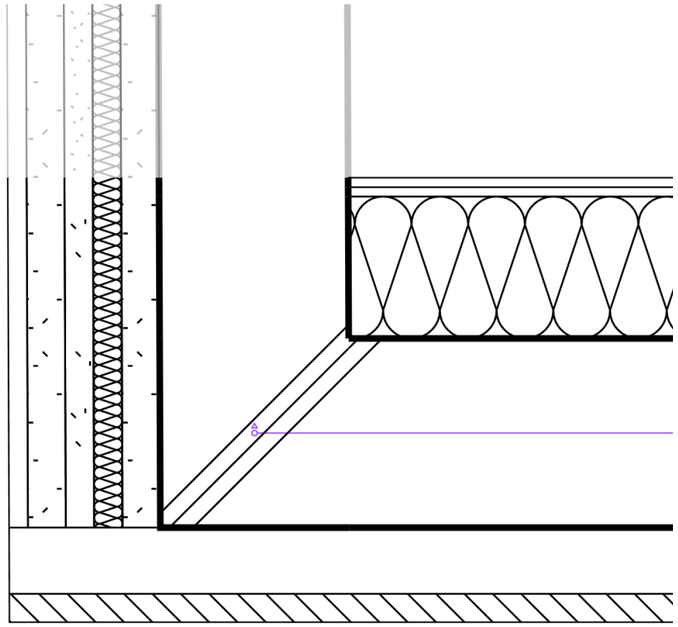
When two sections intersect at a joint, the material priorities are determined by the Easy Join Detail. Selecting a joint and clicking on the overlapping area allows users to reverse the order of priority between the two materials with minimal effort.



Easy Join two or more sections



Cliick the overlapping area



Joint priority is reversed

Shoppertoire

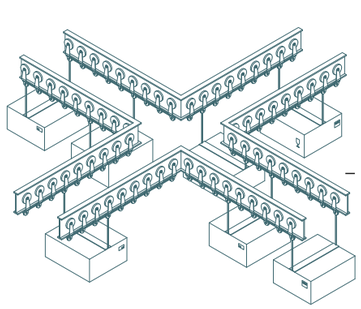
Date	*Spring, 2020*
Type	*Undergraduate Thesis Project*
Keywords	*Modular* *Logistics* *Self-governing architecture* *Market*
Tutor	*John Hong*

Shoppertoire envisions a future of self-governing architecture in which infrastructure serves as a platform for adaptable typologies, particularly suited to commercial programs. Located in the traditional marketplace of Namdaemun in central Seoul, this proposal introduces a flexible system of commerce, distribution, and storage that can rapidly expand, shrink, or self-correct to meet evolving demands. A network of conveyor belts, modular programs, and transparent architectural elements forms the core of the project, reflecting changing retail practices and embracing user agency in shaping the environment. Inspired by Yona Friedman's view of architecture as a "catalogue of repertoires," the design encourages occupants to configure and reconfigure the space according to their needs and aspirations, ensuring continual adaptability and growth.

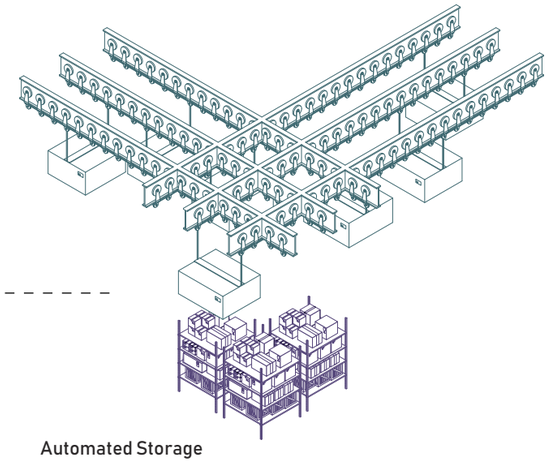




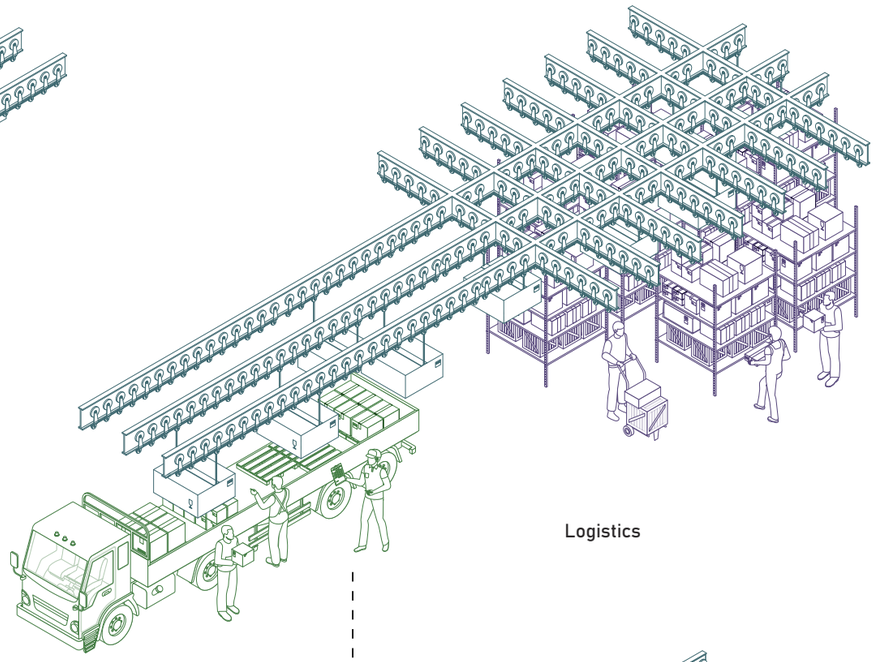
Elemental Repertoire



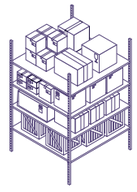
Conveyor Belt



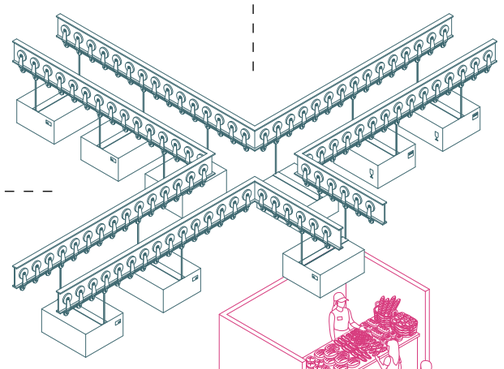
Automated Storage



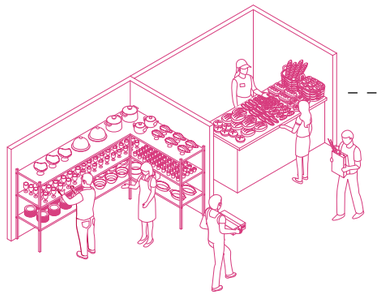
Logistics



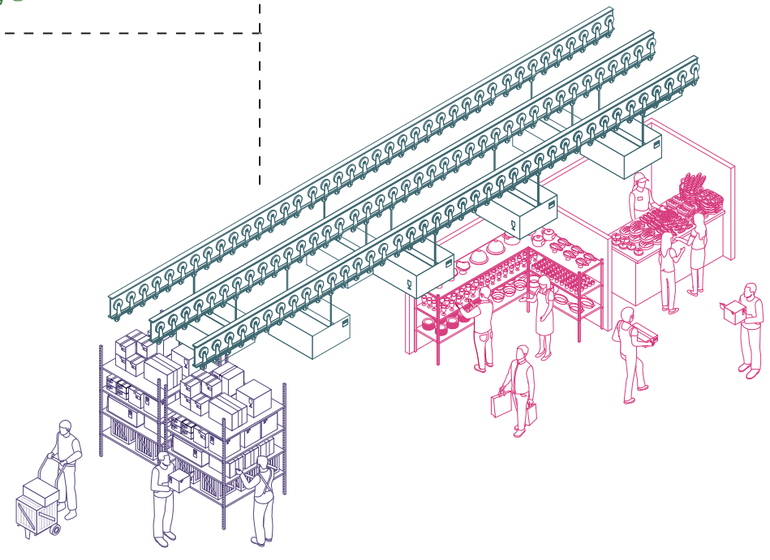
Storage Unit



Shop Delivery



Shop Unit



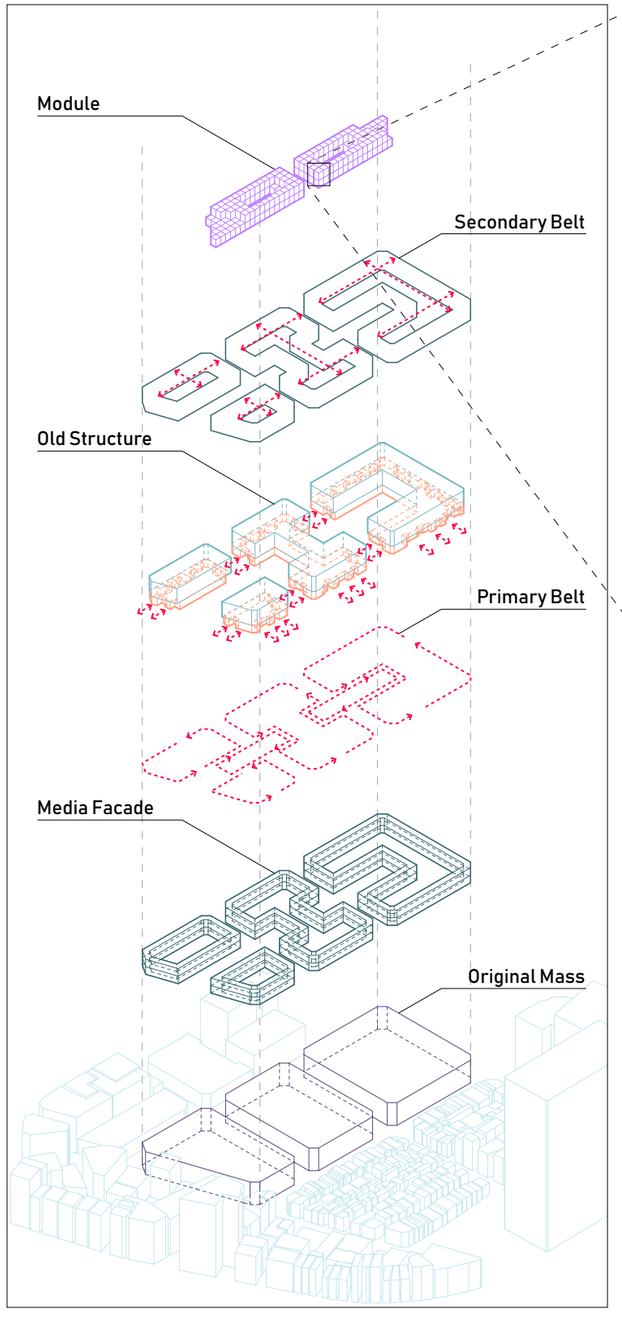
Shared Storage

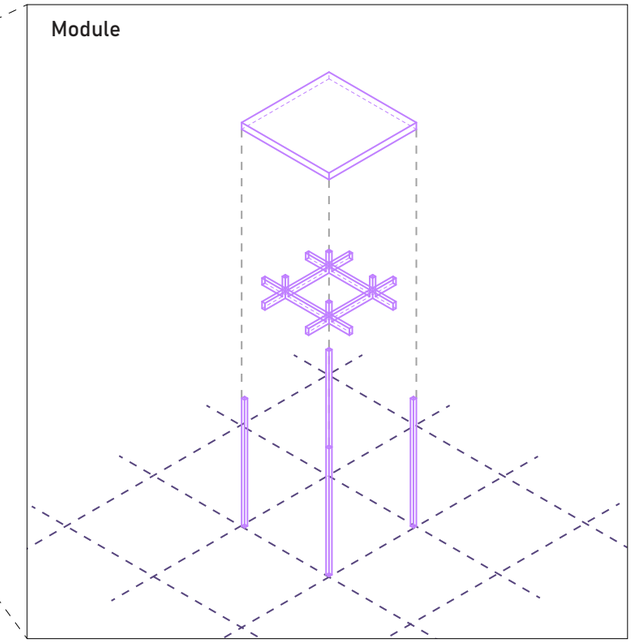
Elements of repertoires create different scenarios when combined with others.

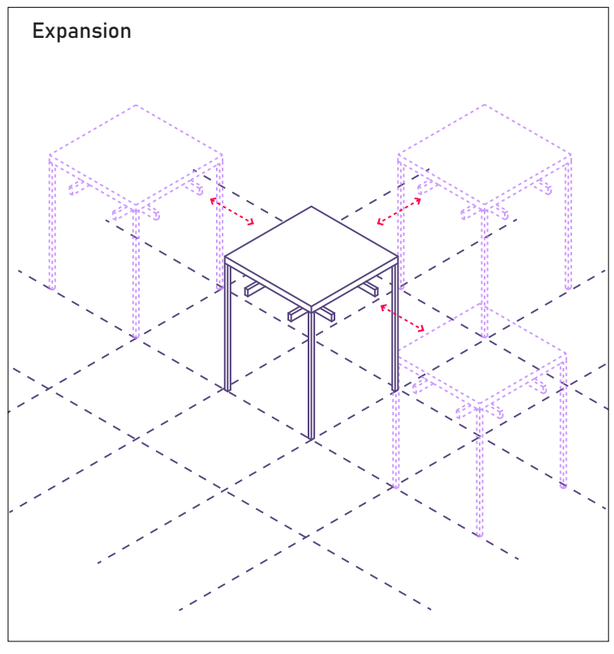
The fundamental building blocks of Shoppertoire are conveyor belts, storage units, and shops. Conveyor belts function as the primary infrastructure, transporting merchandise seamlessly across multiple levels and connecting various fragments of the existing market. This organized flow of goods allows for rapid distribution and efficient restocking processes, reflecting contemporary logistics operations.

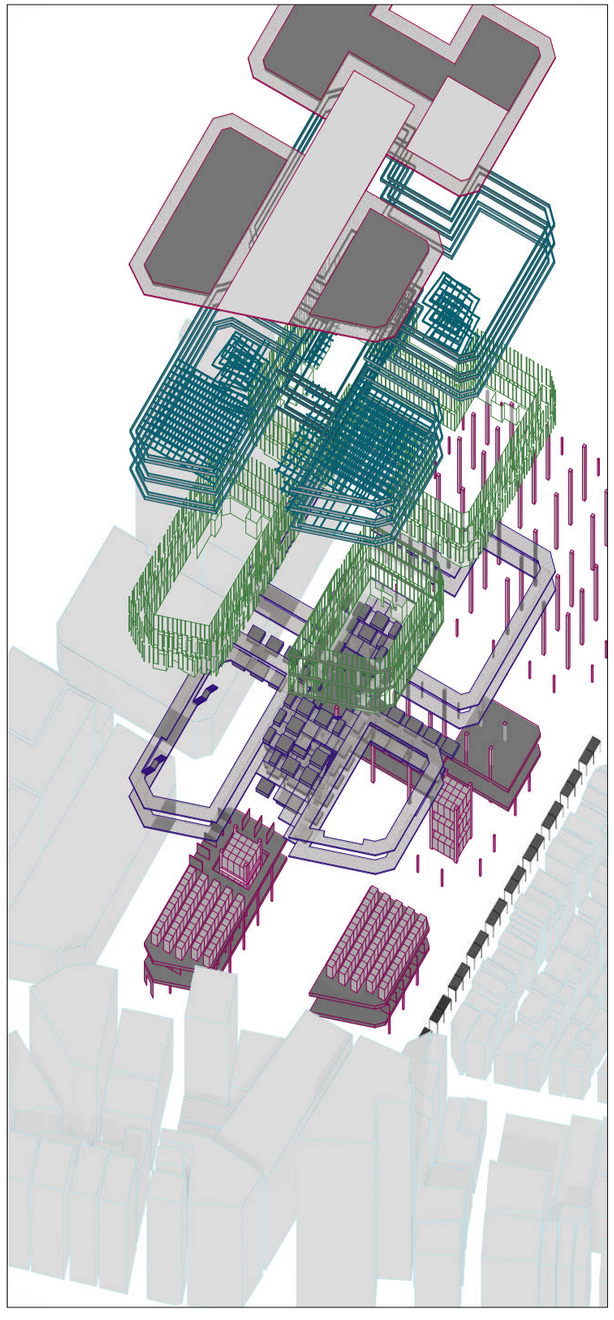
Storage units integrate directly with the belt system, ensuring an optimal balance between available inventory and immediate consumer demand. While these units serve a single purpose when aggregated (such as bulk warehousing), their flexibility increases when linked with the shops. The shops themselves are shaped by modular, transparent enclosures that showcase products and activities within, while media façades puncture the cladding to enhance visibility and marketing potential. By optimizing the size and placement of these symbolic openings, the design aims to encourage open interaction between merchants and patrons, ultimately breathing new life into the market.

Modular Intervention









In addressing the dense spatial conditions of Namdaemun Market—where narrow 3-meter-wide stalls are jammed side by side—Shoppertoire proposes a reorganized framework that weaves its conveyor belt infrastructure around and through existing structures. This primary belt loops around the boundary, while secondary pathways spread inward like a web, connecting individual programs to the broader system. The voids between existing fragments provide space for inserting modular units that expand and reshape the market's footprint, responding to user-driven demands for additional or reconfigured program space.

The resulting network of decks, ramps, and towers creates multiple layers of circulation. Vertical conveyor belts move goods from a loading dock located adjacent to major roads up through stacked towers of shops and warehouses, streamlining deliveries and shipments in a centralized hub. Surrounding programs can link onto these main vertical pathways, reducing congestion and improving accessibility. By situating the flexible modules within leftover or underutilized areas, the design prompts new patterns of interaction and commerce to emerge, ensuring that each intervention complements, rather than replaces, the traditional market setting.

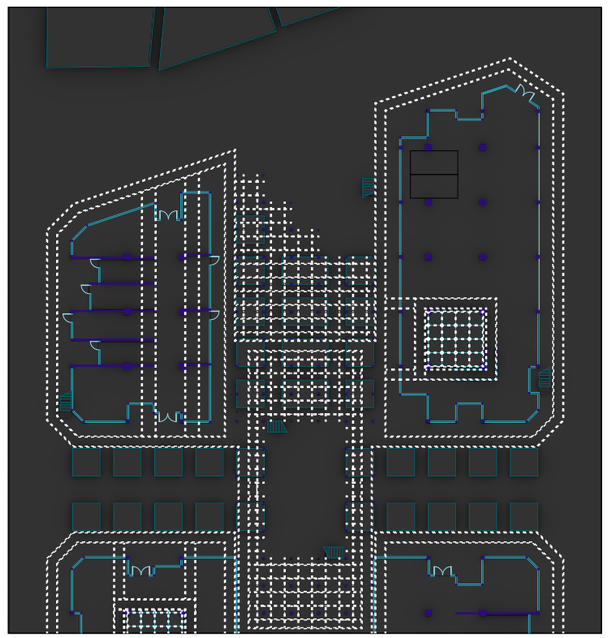
Plan Level 1



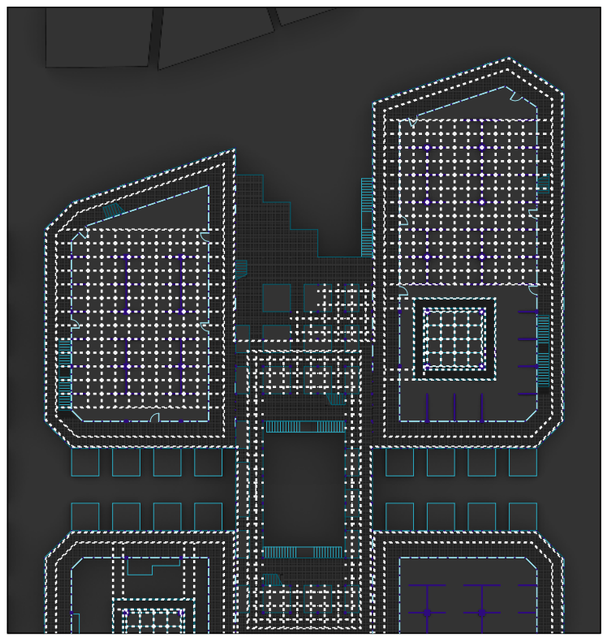
Plan Level 2



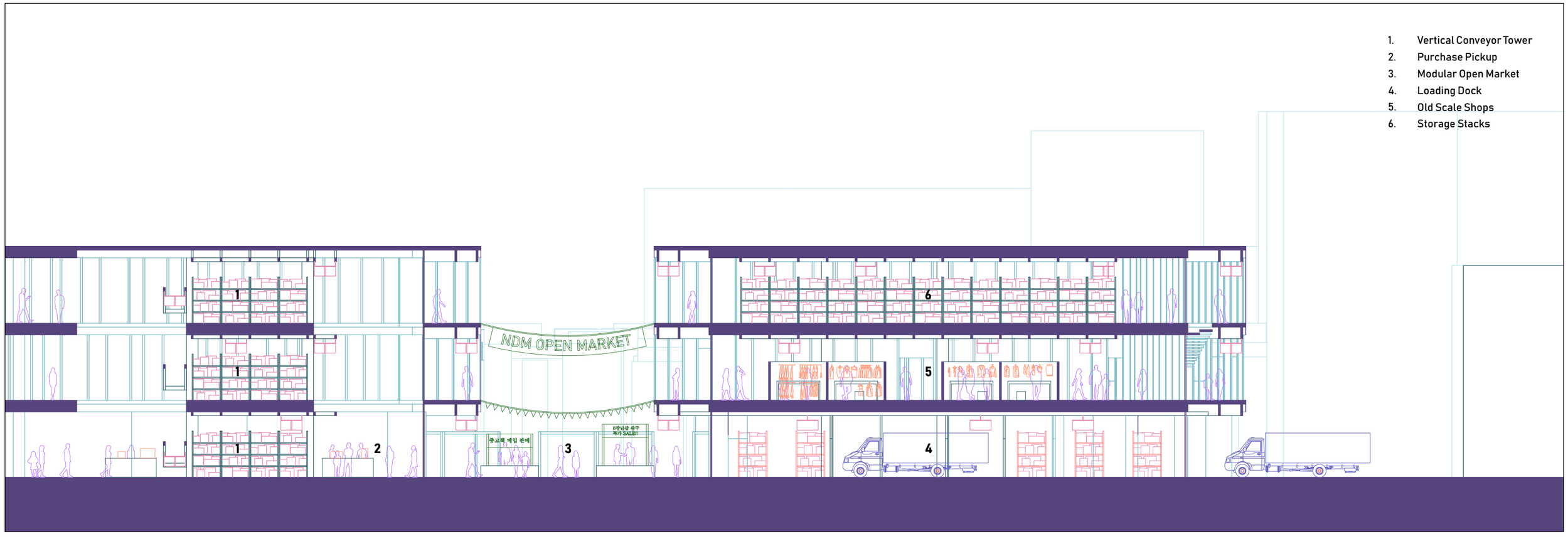
Infrastructure Level 1



Infrastructure Level 2



SN Section



Scenarios

Shoppertoire's adaptability manifests in numerous scenarios, reflecting the varying needs of merchants, shoppers, and logistics providers. When conveyor belts, storage units, and shops remain isolated, each element can function efficiently within its individual program. However, the most transformative possibilities arise when these elements are aggregated or interconnected. Multiple shops can expand vertically or horizontally, forming dynamic clusters that support additional retail zones, pop-up events, or collaborative market experiences.

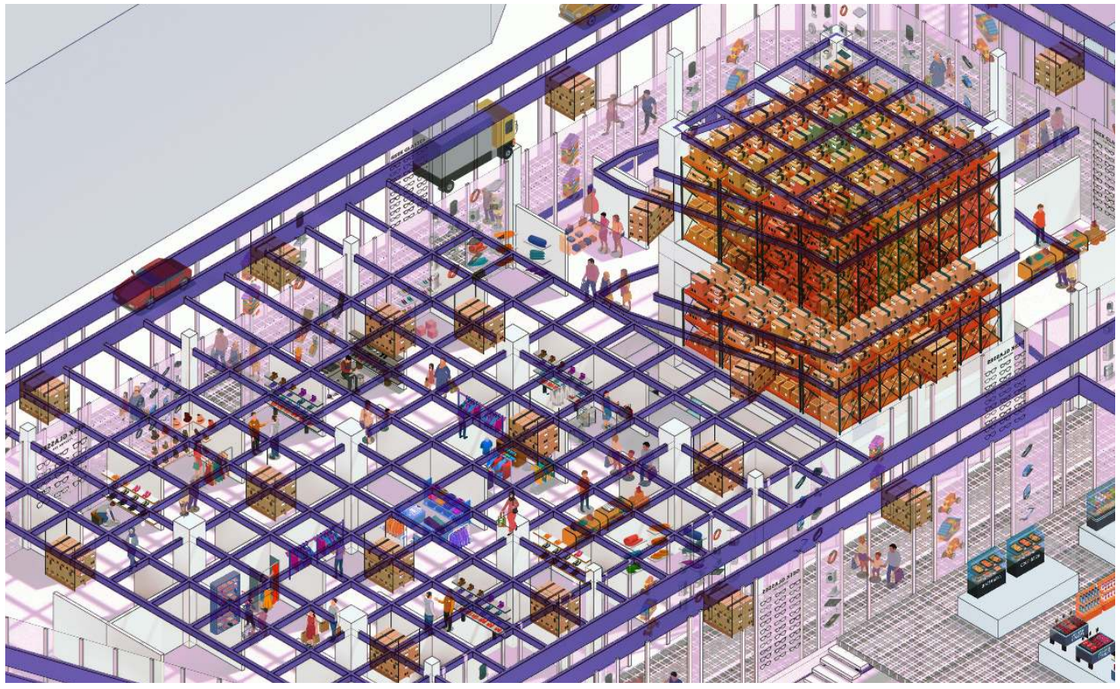
The layered belts and modules also accommodate digital integration through media façades and online order processing, allowing merchants to reach both local and virtual audiences. Storage hubs can scale in tandem with temporary expansions—such as seasonal markets or promotional events—and the belt system's capacity can be adjusted to manage increased flow. Across all variations, Shoppertoire's repertoire of elements and infrastructural support aligns with the evolving nature of commerce, encouraging inhabitants to take an active role in shaping and reshaping their built environment over time.

DELIVERY - LOAD - BELT



The loading dock is situated by the big street where shipment to the shops and from the shops to online customers are made. The vertical stack tower carries the merchandise from the loading dock to the shops and warehouses.

SHOP - BELT - STORAGE



The primary, secondary and vertical conveyor belts connect the small scale shops to the vertical stack tower and the surrounding system. The screen façade provides visual and informational transparency by indicating which items are being sold in each area.

MODULE - CIRCULATION



The modules expand the infrastructure and the program out into the in-between voids, and at the same time connects the fragments and provide different kinds of circulation.

Scenarios









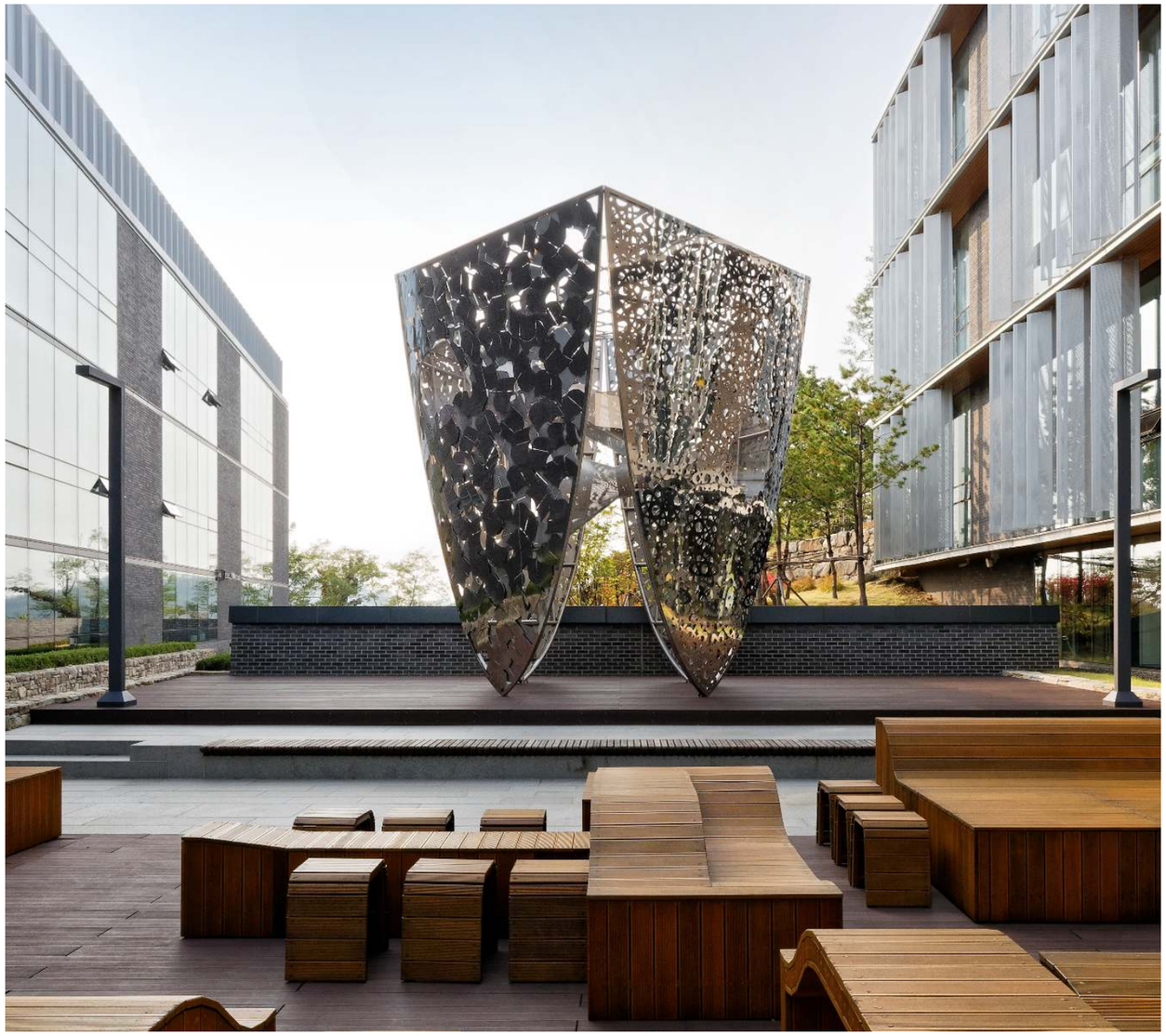


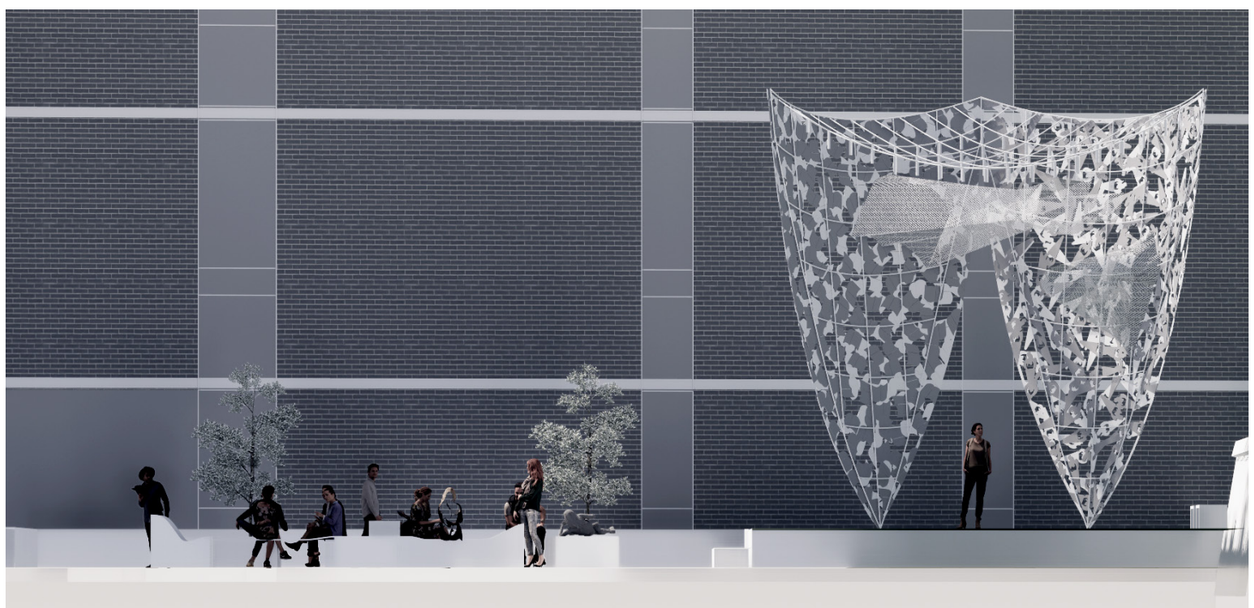


SFS Monument

Date	*Spring, 2019*
Type	*Professional Internship*
Keywords	*Parametric Design* *Modular Furniture* *Built Work*
Office	*Project: Architecture*
Role	*Parametric Design;* *Construction Documentation*

This project transforms the courtyard of Seoul Foreign School into a dynamic environment centered around a pavilion-like monument and a series of modular furniture. The monument incorporates four shields that symbolize the school emblem's core icons—the book, the torch, the atom, and the cross—linked by wormhole-like mesh tunnels and topped with an interactive LED lighting system. Scattered across the courtyard, modular furnishings can be combined and rearranged to accommodate daily student life, lectures, stage performances, or casual gatherings, ensuring the outdoor space remains flexible and engaging.





Monument

The monument in the courtyard represents the four symbols of the school emblem: the book, the torch, the atom, and the cross. These symbols are expressed through four stainless steel shields, each formed by two overlaid layers supported by a tubular frame.

The surface of every shield features a randomly scattered pattern of its respective emblem, optimized through Grasshopper to maintain sufficient transparency while preserving the rigidity of the steel plate. Variables such as size, density, and randomness determine the final aesthetic outcome, striking a balance between openness and stability.

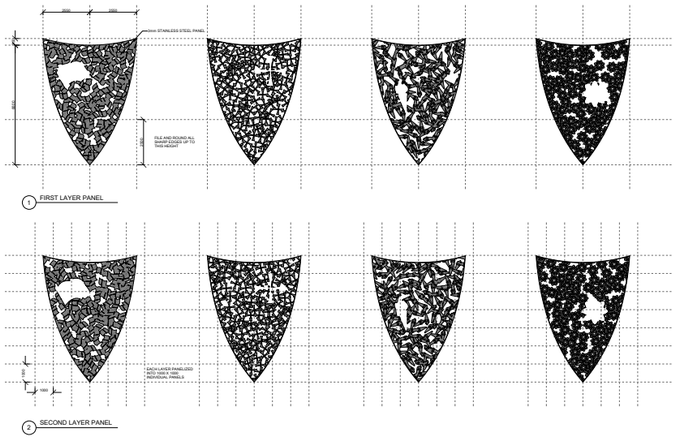
Connecting opposite shields are wormhole-like mesh tunnels that serve as structural support and provide visual continuity among the four icons. Each mesh tunnel's cross section gradually transitions from one emblem to another, creating a cohesive sense of flow.

Positioned at the top of the monument, a hanging interactive LED lighting system reacts to passersby, introducing a playful layer of interactivity. Installed on a grass-covered section of the courtyard, the monument stands as an inviting focal point, encouraging spontaneous exploration and social encounters.

Shield Pattern Close-up



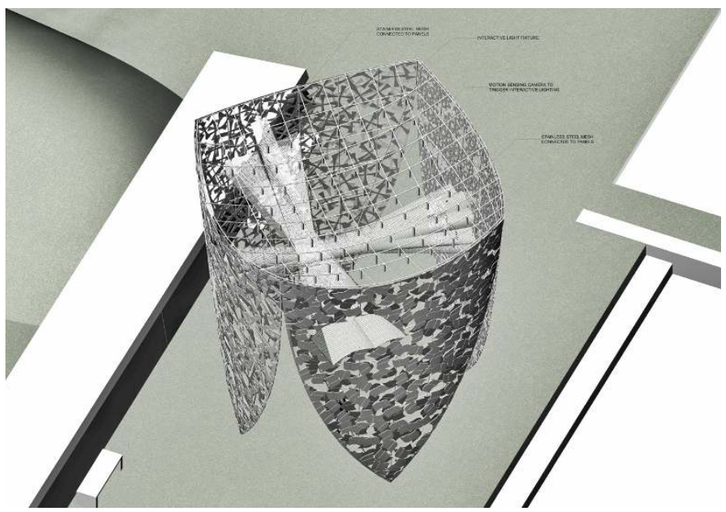
Shield Plate Detail



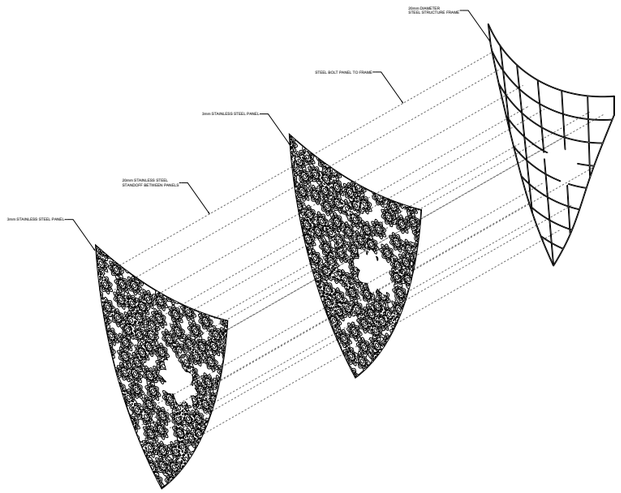
Inner View



Axonometric View



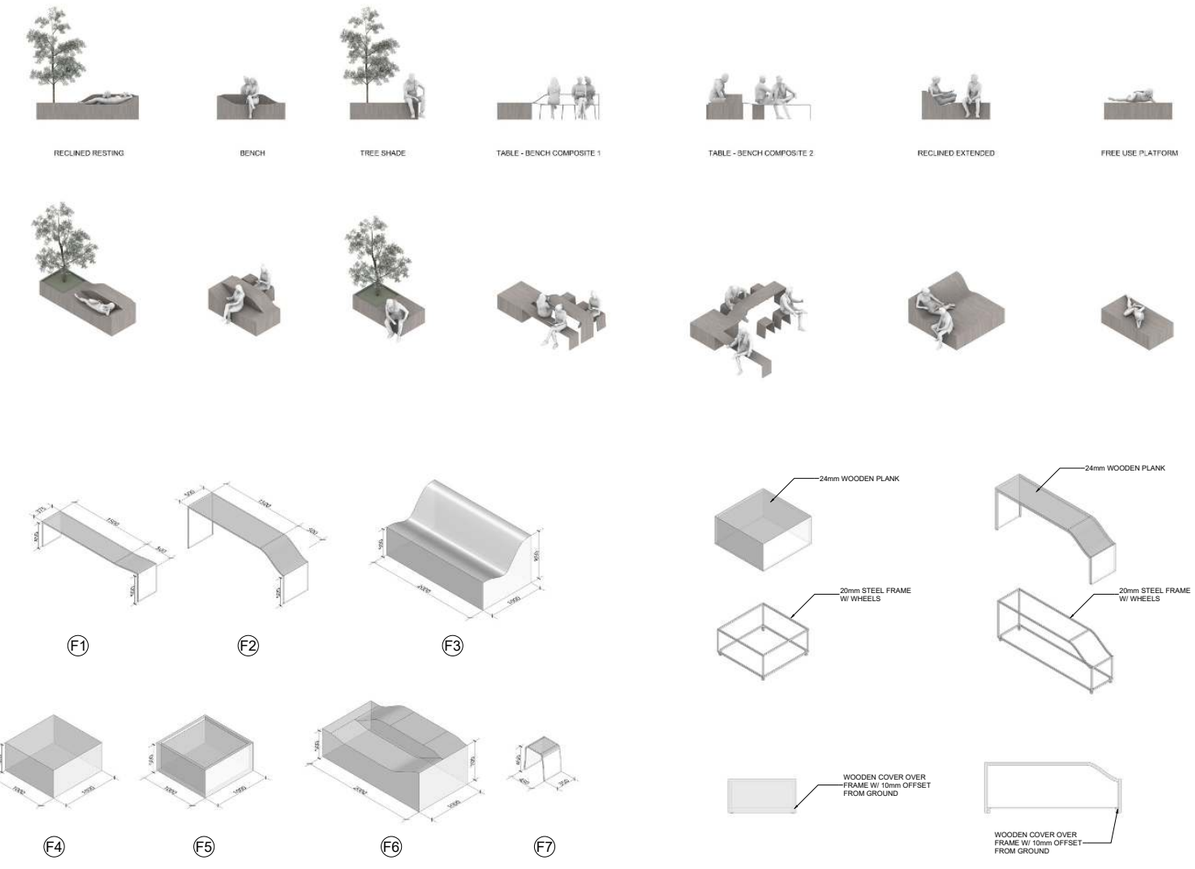
Shield Composition

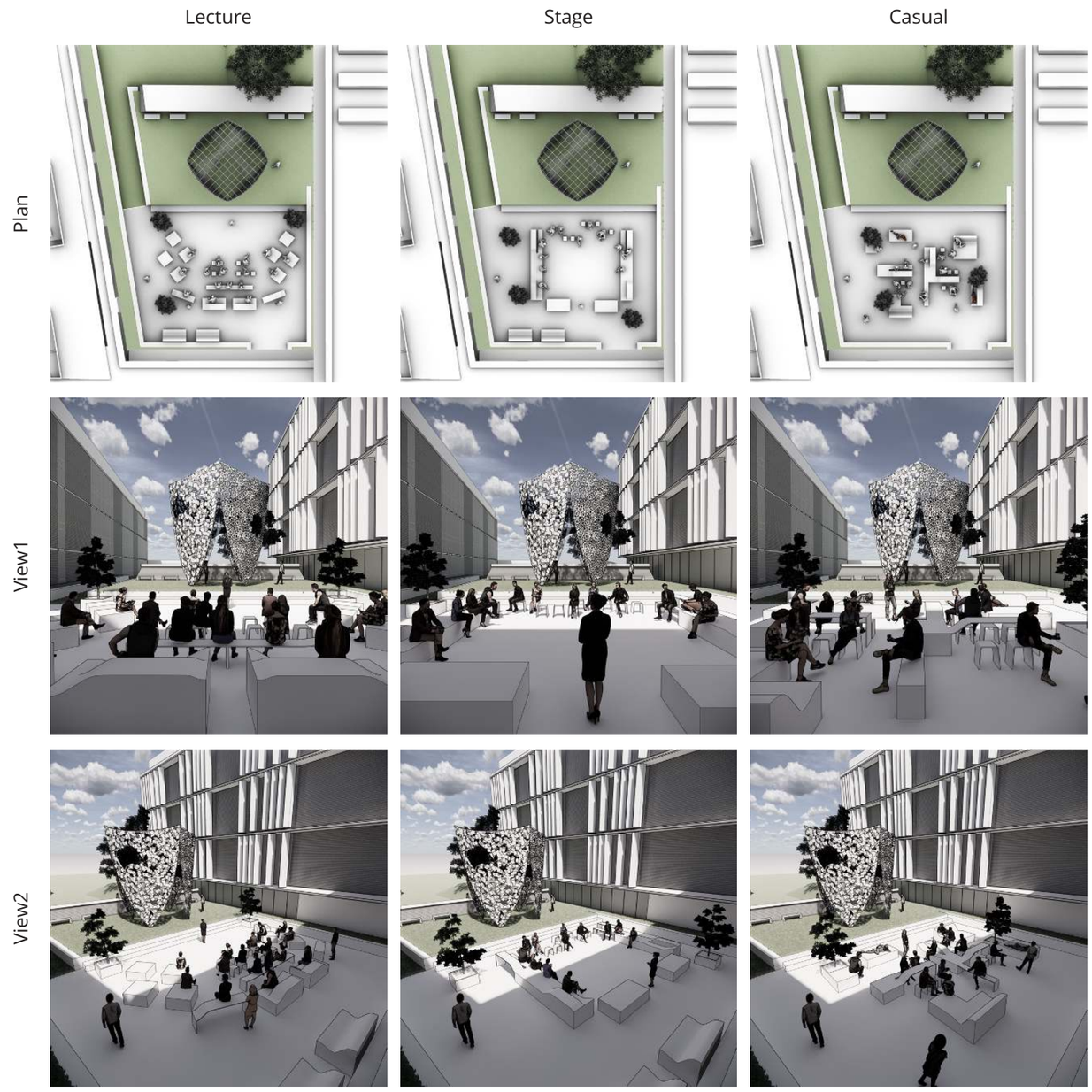


Modular Furniture

On the adjacent paved area, modular furniture enhances the courtyard's versatility. These pieces—tables, chairs, reclining benches, platforms, and planted elements—are finished with wooden planks supported by steel frames, some of which are equipped with wheels or other movable components for easy rearrangement.

Designed to suit a wide range of activities, the furniture can be combined in various configurations to support anything from informal group discussions and study sessions to organized lectures or stage performances. By placing these rearrangeable units alongside the monument, the courtyard is activated as a multi-purpose space where students and faculty can adapt the layout to their needs.

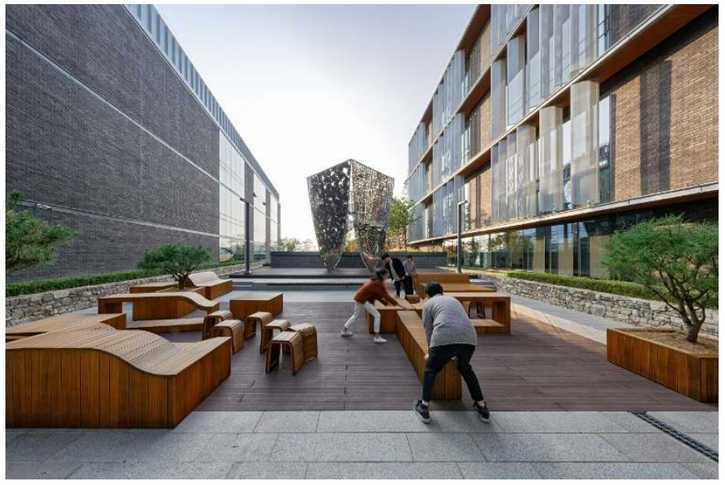




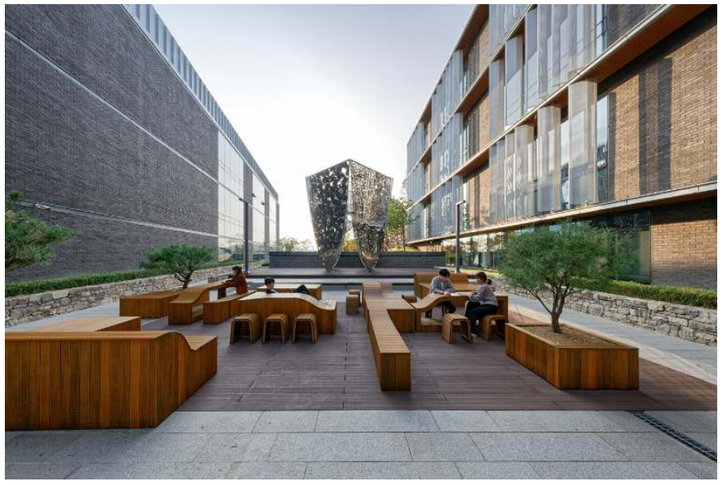
Compositions



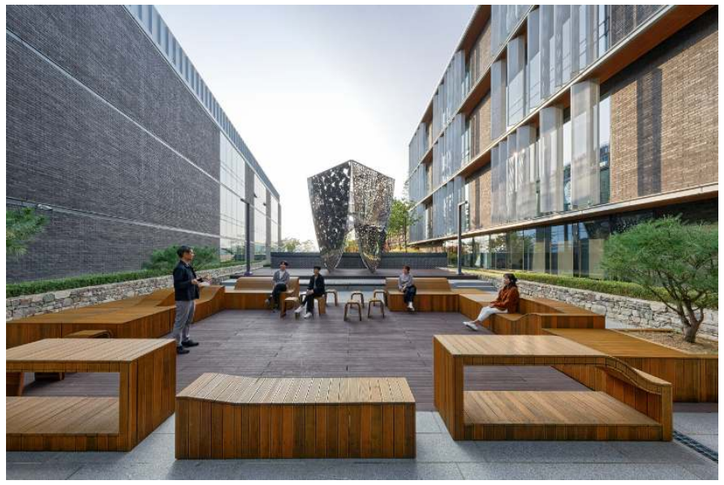
Lecture



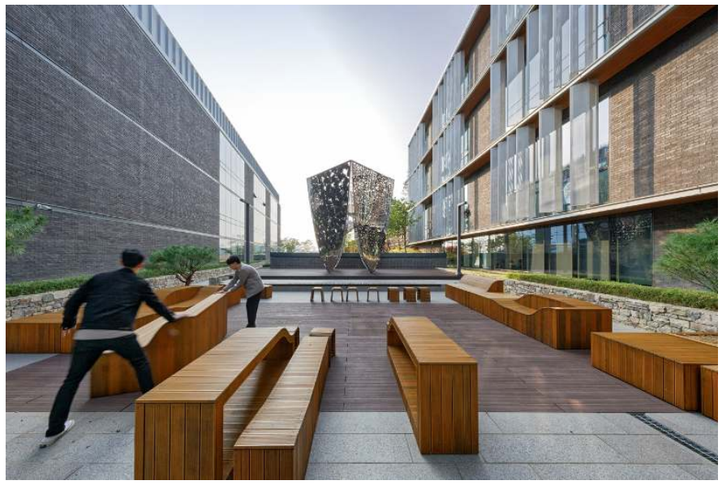
Stage



Casual



Transition



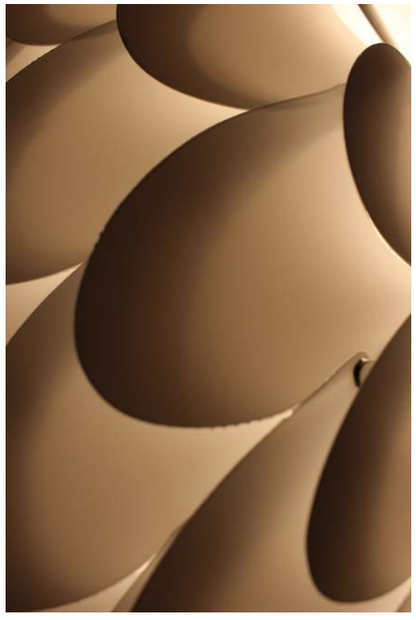
Transition

MorphoLux + Manta

Date	*Fall, 2016*
Type	*Undergraduate Digital Design Studio*
Keywords	*Parametric Design* *Fabrication*
Supervisor	*Hangman Zho*
Collaborators	*Jaejin Lee - MorphoLux* *Daye Kim, Kiwon Jeon - Manta*

These two projects were developed as part of the digital design studio class at SNU, highlighting the potential of parametric design and user-driven customization. MorphoLux and Manta both rely on Grasshopper to generate a range of effects by altering the parametric conditions from which their forms are derived, showcasing how digital scripting can streamline fabrication and help produce complex geometries at varying scales—from personal lamp shades to large-scale installations.









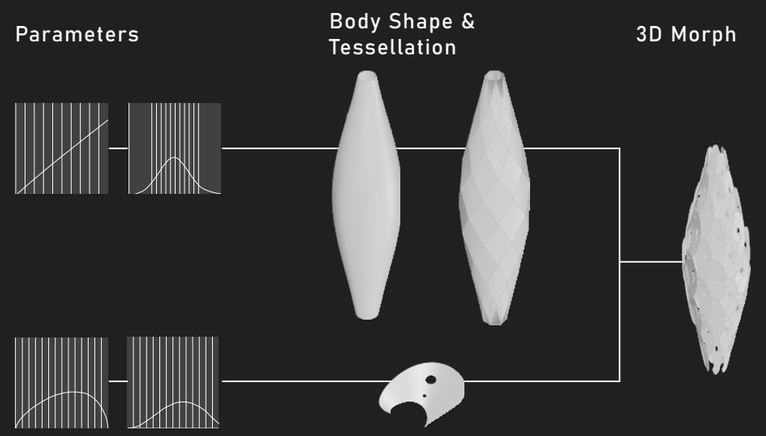
MorphoLux

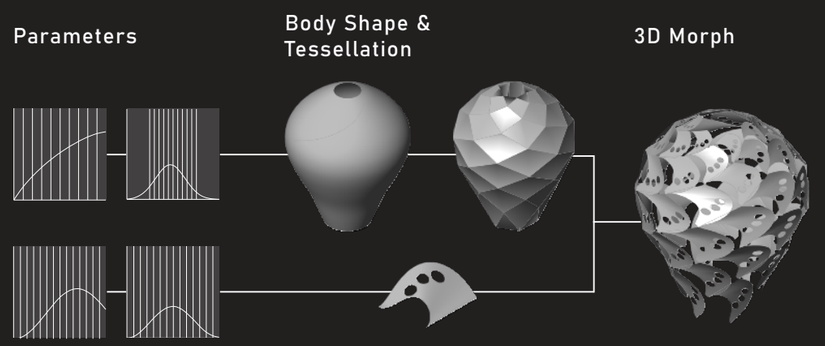
MorphoLux is a series of lamp shades created as an introductory project for exploring form-making through Grasshopper. The primary goal was to provide a versatile design mechanism, where users have the freedom to customize the final product to their liking by downloading and modifying an open-source algorithm. By altering the parametric settings—such as the shape, proportions, or tessellation density—users can achieve a wide spectrum of aesthetics, ranging from subtle and minimal to more intricate and sculptural.

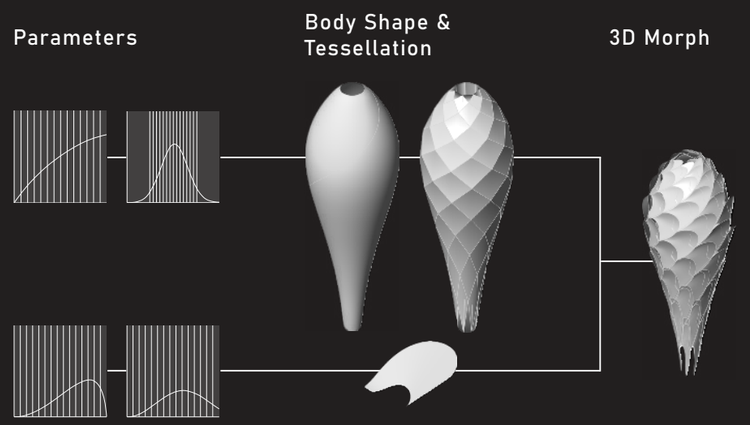
This project is deliberately set up as a DIY process, making it accessible to individuals with varying levels of design and fabrication experience. MorphoLux is designed to be constructed using basic materials like cardboard or plastic sheets, which can then be fastened together with simple clips or other readily available fasteners. Because the parts can be printed on standard home or office equipment, assembling each unique lamp shade becomes a hands-on exploration of computational design principles.



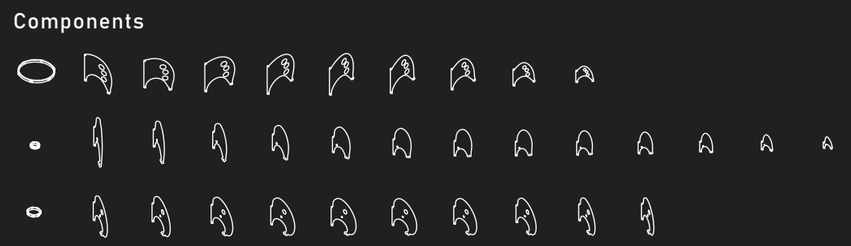








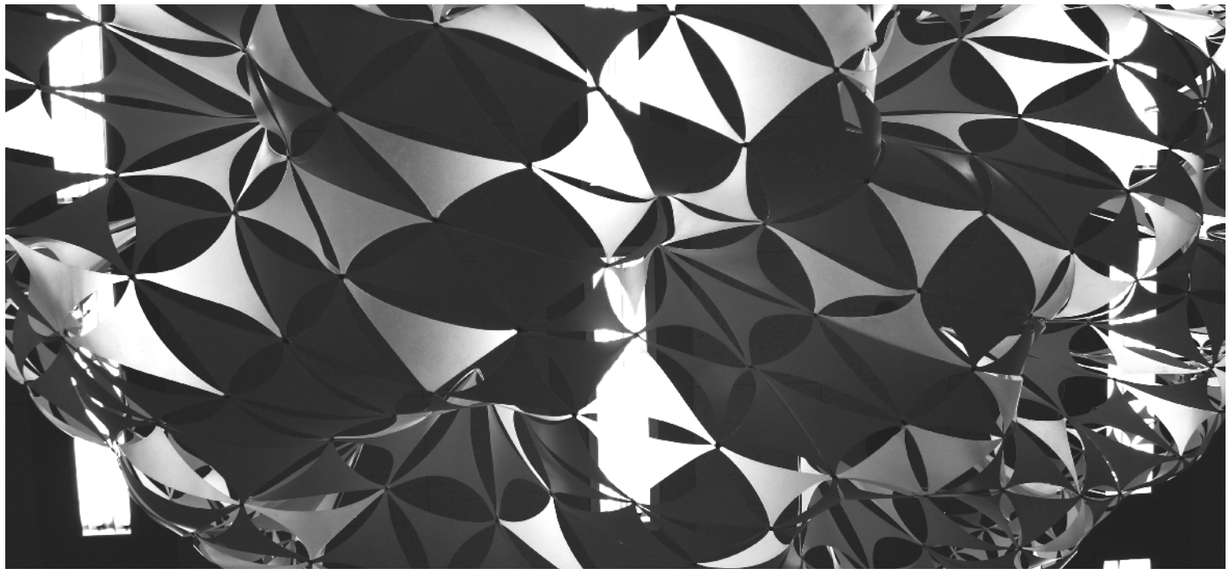




Manta

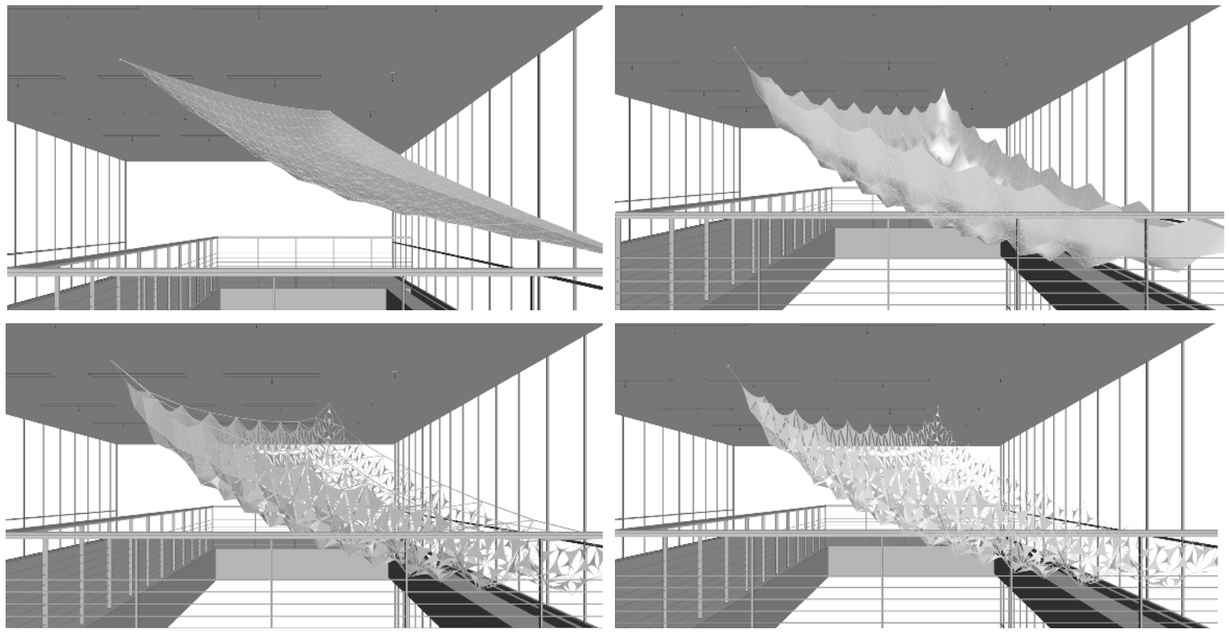
Manta was selected in an in-class competition to be installed at the SNU DAAE hall, serving as a pavilion-like spatial intervention. Its initial design began with a single large triangle, which was then broken down into smaller triangles to form a tessellated surface. Anchor points were chosen, and a series of simulated forces pulling towards the ground was applied, allowing the piece to naturally assume its final curved and flowing form. This computational step ensured that the resulting mesh not only looked visually dynamic but also maintained structural logic.

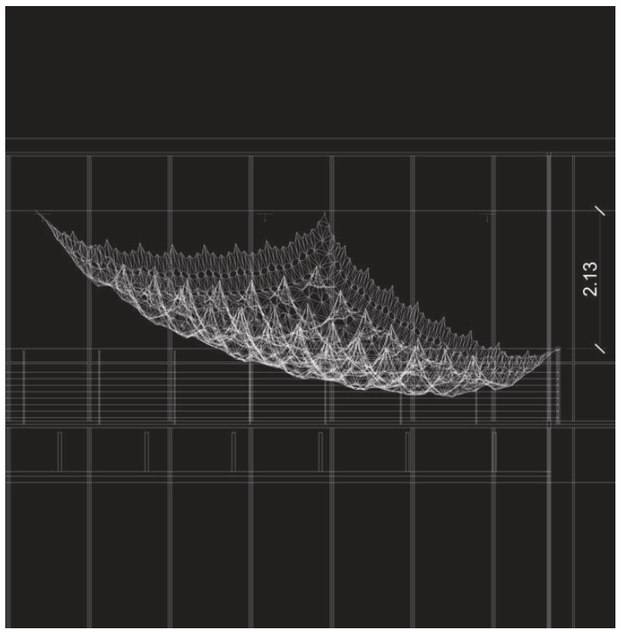
One of the key challenges in bringing Manta to life was the sheer number of individualized units—hundreds of them—each of which had to be precisely placed and assembled. Python scripting was used to manage the fabrication workflow, helping to organize the necessary data and ensure that every piece would fit seamlessly into the larger structure. Once the tessellated modules were fabricated, they were suspended from designated vertices using cables, which provide critical support and prevent excessive swaying or deformation. In addition, punctures were introduced between the overlapping layers, allowing different amounts of light to pass through. This interplay of light and shadow animates the space below, creating an immersive environment that shifts throughout the day.

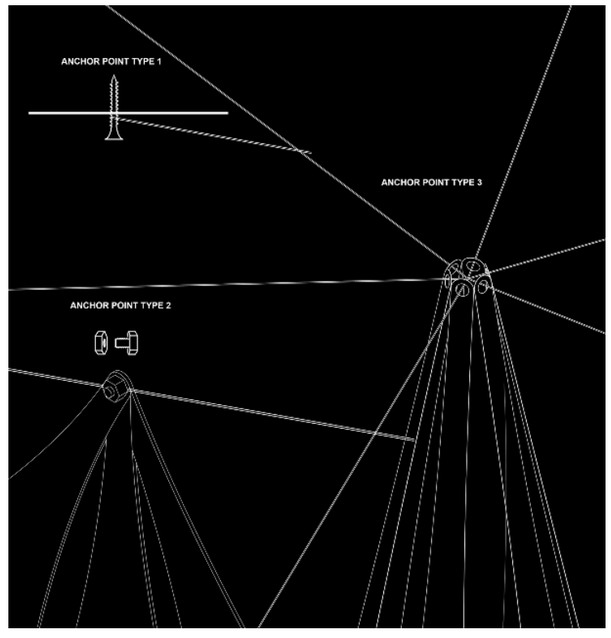


Elevation

Detail







Manta_Tesselation: Tesselated components are shrunk according to position

